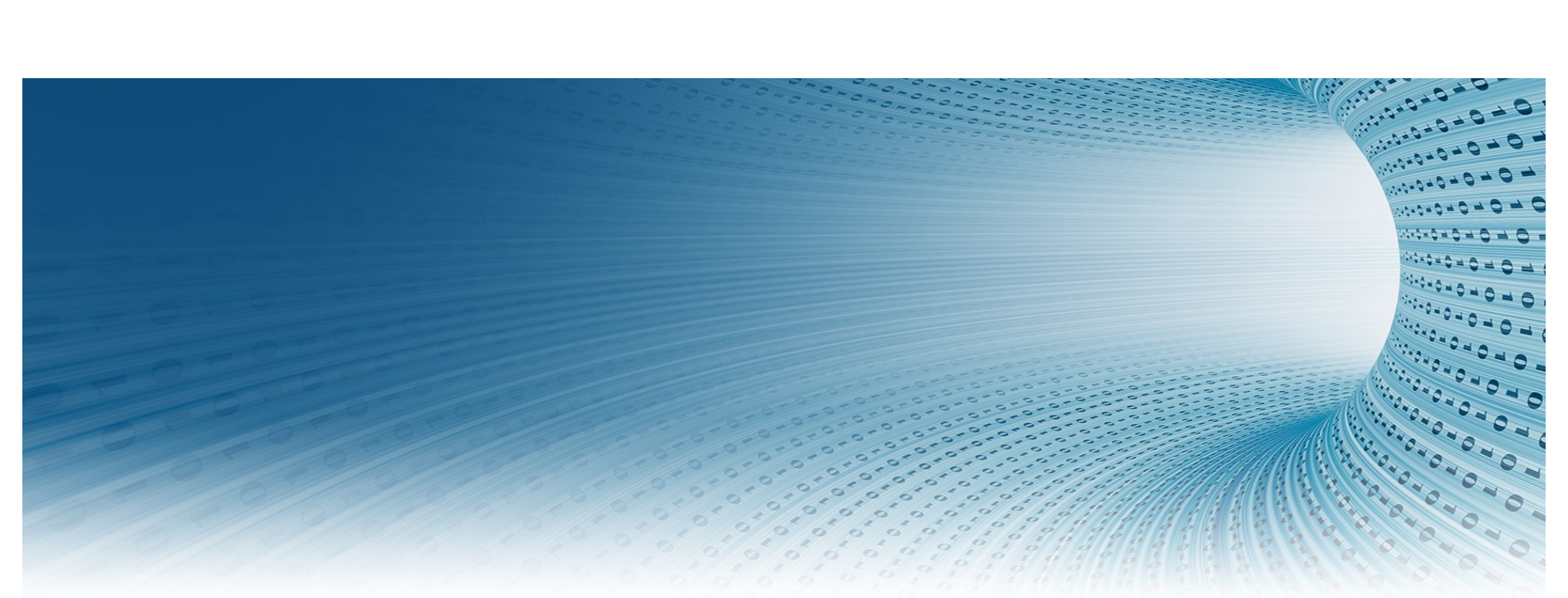


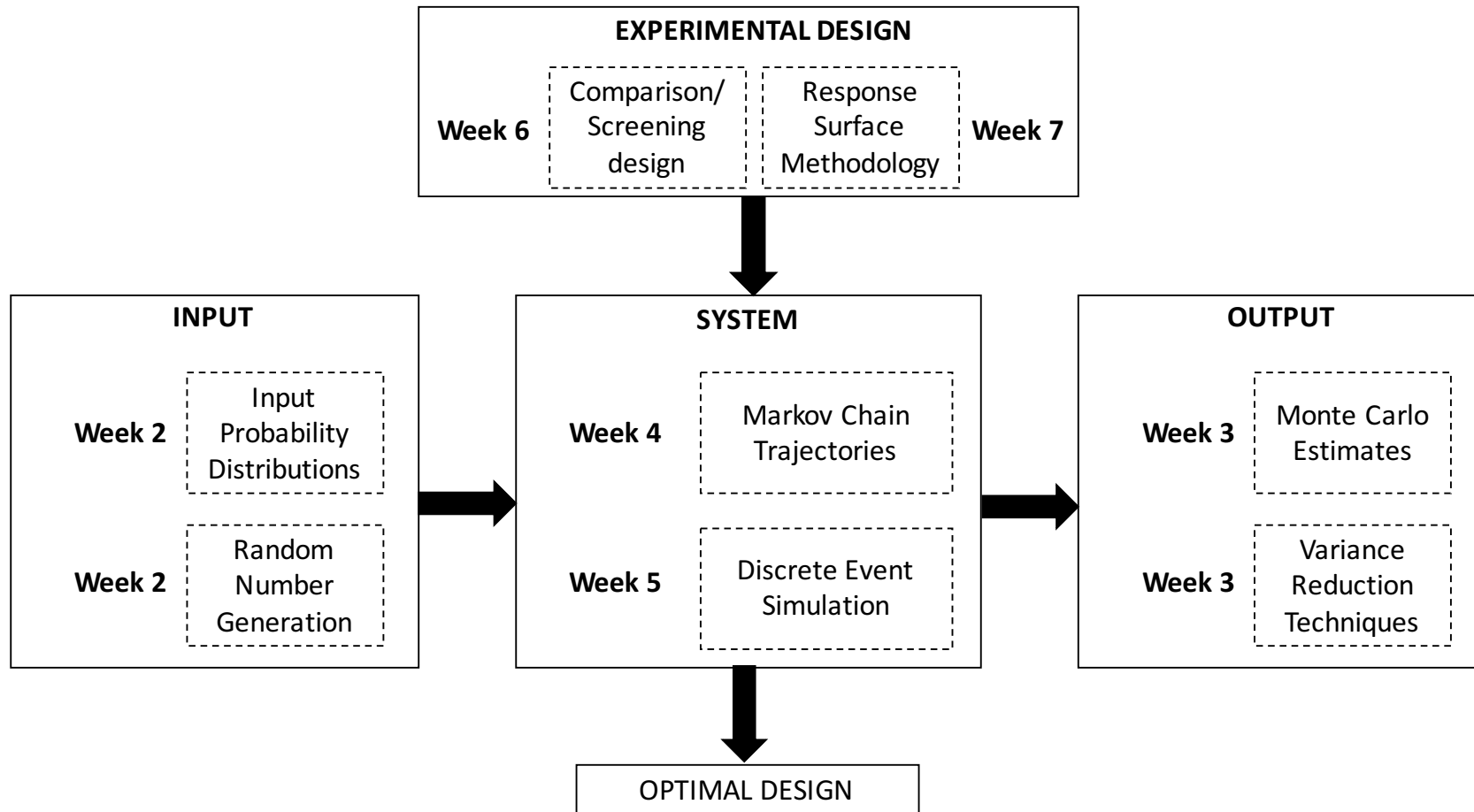


Experimental Design and Optimisation

Theory and formulation

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Faculty of Economics and Business Administration
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9000 Gent

Course Schedule



Outline

- Introduction
- Experimental design
- Comparison between design alternatives
 - Comparison between two design alternatives
 - Comparison between more than two design alternatives
 - Ranking and selection of design alternatives
- Screening design
- Response surface design: Optimal design of the system
 - Stochastic approximation
 - Alternative Heuristic optimisation: Simulated annealing

Introduction

- Design of a system
 - Formally, a design α can be determined completely by one or more parameters $(v_1, v_2, \dots, v_k)$, e.g.
 - Maximum capacity of inventory
 - Economic order quantity
 - Number, speed or quality of machine, . . .
 - Replenishment policy, priority policy
 - . . .
 - Design space: set $\mathcal{A}$ of all feasible values α . Each design has a relevant cost $J(\alpha)$ relating to aspects of
 - Implementation
 - Desirability, acceptability, . . .
 - Performance => Evaluated by Discrete Event Simulation

Experimental design

- Experimental design terminology

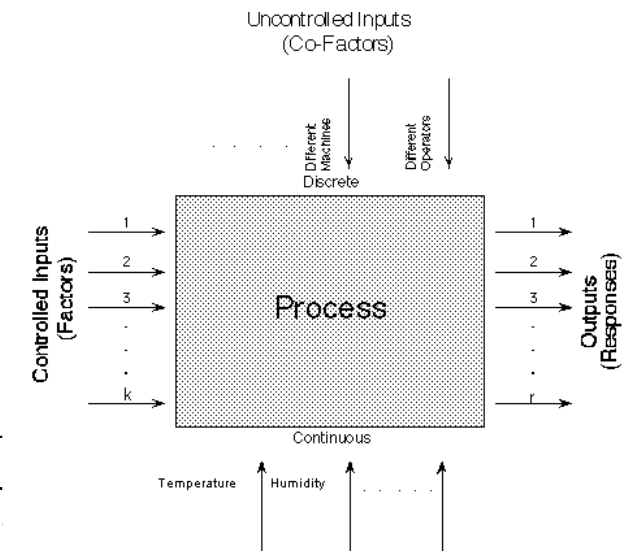
- Input parameters and structural assumptions composing a model are called factors

- Quantitative (or numerical) vs qualitative (or categorical) factors
- Controllable (focus) vs uncontrollable factors

- Output performance measures are called responses

- Example

System	Possible factors	Quantitative or qualitative?	Possible responses
Inventory system	Reorder point	Quantitative	Average cost per month, average number of items in inventory
	Order-up-to level	Quantitative	
Manufacturing system	Number of machines	Quantitative	Average time in system, throughput, average queue size
	Number of forklifts	Quantitative	
	Forklift dispatching rule	Qualitative	
Communications network	Number of nodes	Quantitative	Average end-to-end delay, throughput
	Number of links	Quantitative	
	Routing policy	Qualitative	



Experimental design

- The choice of an experimental design depends on
 - the objectives of the experiment
 - Comparative objective: The goal of the *comparative design* experiment is to make a conclusion about one a-priori important factor and to determine whether or not the factor is significant, i.e., whether or not there is a significant change in the response for different levels of that factor.
 - Screening objective: The purpose of the *main effects design* experiment is to select or *screen out* the few important main effects from the many less important ones.
 - Response Surface (method) objective: The *response surface method (RSM) design* experiment allows to estimate interaction and even quadratic effects (e.g. via regression analysis). This gives an idea of the (local) shape/mathematical function of the response surface, which can be optimised.
 - the number of factors to be investigated

Experimental design

- The choice of an experimental design

Number of factors	Comparative Objective	Screening objective	Response Surface Objective
1	1-factor completely randomised design	-	-
2-4	Randomised block design	Full or fractional factorial	Central Composite or Box-Behnken
5 or more	Randomised block design	Fractional factorial or Placket Burman	Screen first to reduce number of factors

Reference

Montgomery, D.C., 2013, Design and Analysis of Experiments, 8th Edition, John Wiley, New York

Screening Design

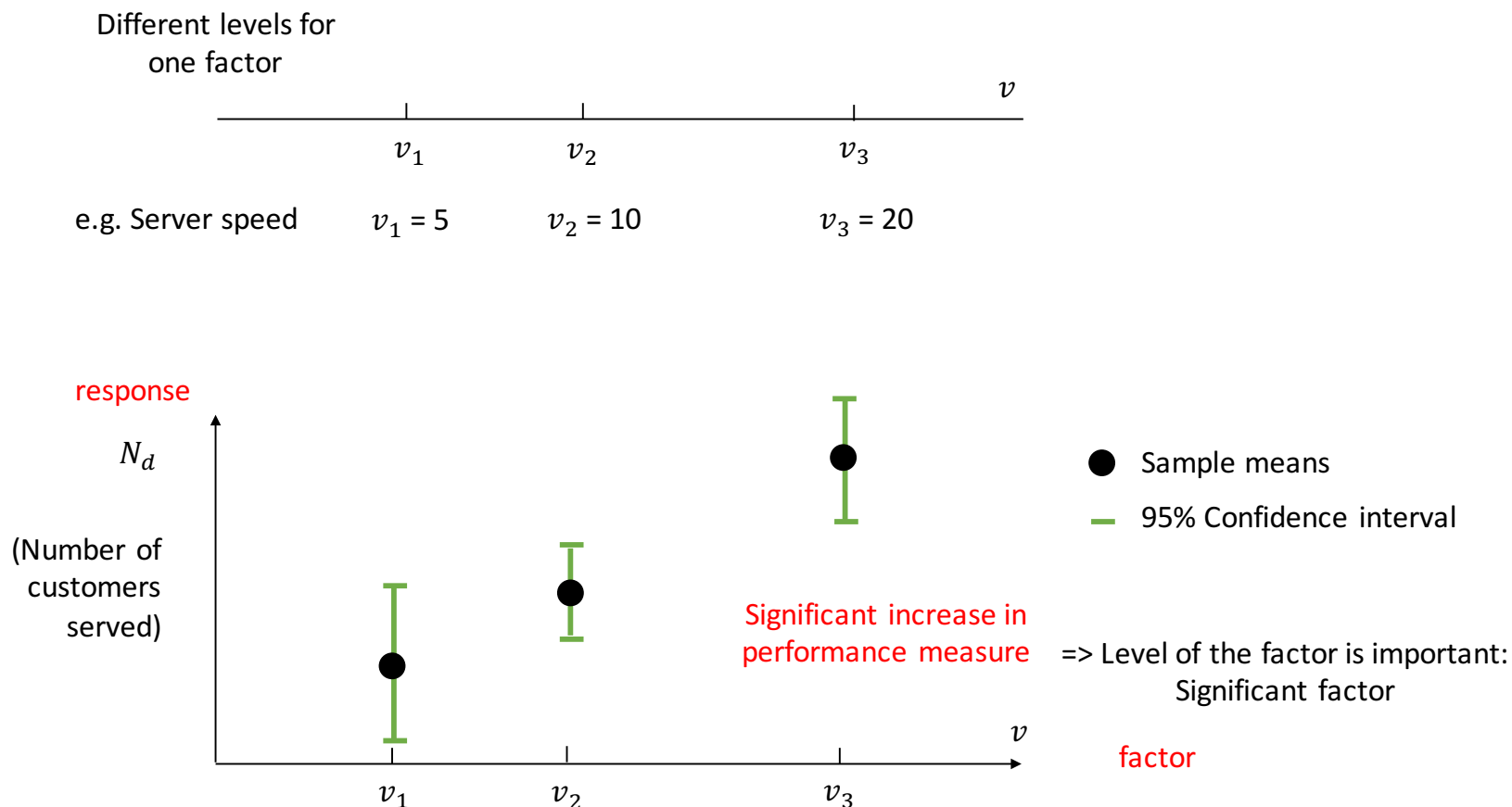
- How to construct relevant designs?
 - Previously we assumed that alternative system configurations were given
 - Research questions
 - Which of the model parameters or structural assumptions impact the performance measure the most or lead to the optimal performance? (*factor screening* or *sensitivity analysis*)
 - How to find relevant alternatives for comparison?
 - Screening design methodology is particularly useful in the early stages of experimentation
 - One-factor-at-a-time approach
 - 2^k factorial design
 - 2^{k-p} fractional factorial design

Screening Design

- Screening design methodology: *One-factor-at-a-time approach*
 - A model with only one factor
 - Run n independent simulations at various values or levels of the factor
 - Calculate confidence intervals for the expected response at each of the factor levels
 - A model with k factors ($k \geq 2$)
 - Determine initial estimate of how each factor affects the response
 - Determine if the factor interacts with other factors, i.e. whether the effect of one factor on the response depends on the levels of the others
 - Method
 - Fix the level of the other $k-1$ factors at some set of values
 - Vary the other factor
 - Examine how the response reacts to changes in the single factor
- => The whole process is repeated to examine each of the other factors, one at a time
- => Inaccurate and inefficient approach, i.e. not all interaction effects are explored or requires too many simulations to investigate interactions

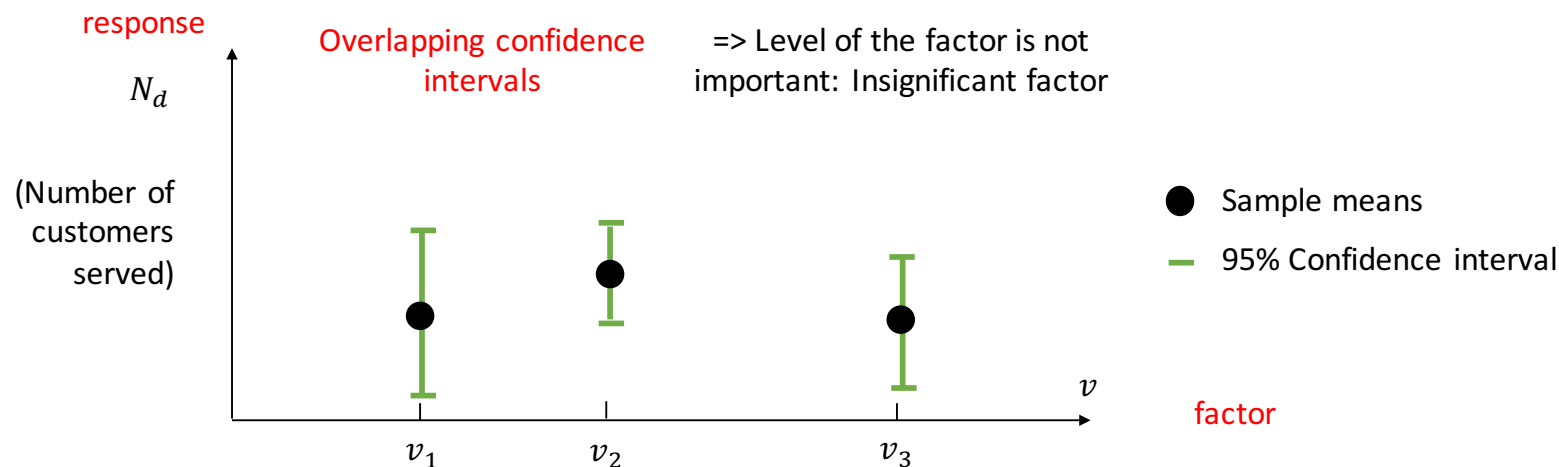
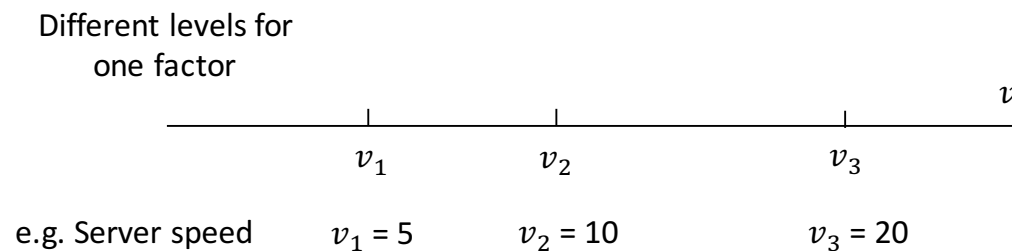
Screening Design

- Screening design methodology: *One-factor-at-a-time approach*
 - A model with only one factor (notes on the blackboard – not in your slides)



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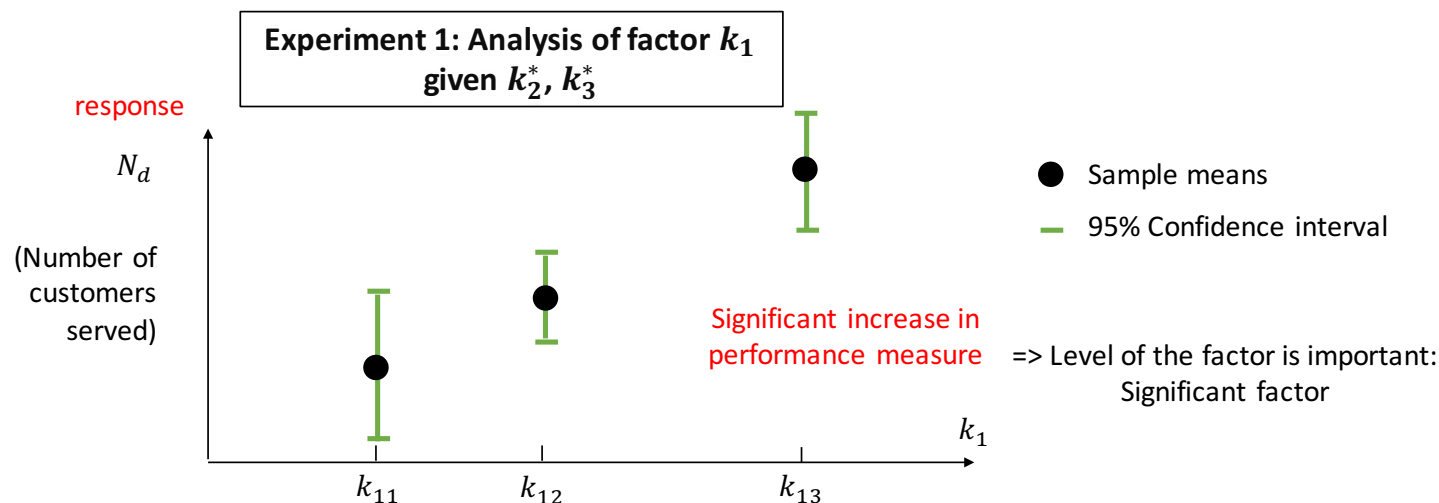
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Screening Design

- Screening design methodology: *One-factor-at-a-time approach*
 - A model with k factors ($k \geq 2$) **(notes on the blackboard – not in your slides)**

Input values	Experiment 1	Experiment 2	Experiment 3
$k_1^* = k_{11}$	k_{11}, k_{12}, k_{13}	$k_1^* = k_{11}$	$k_1^* = k_{11}$
$k_2^* = k_{21}$	$k_2^* = k_{21}$	k_{21}, k_{22}, k_{23}	$k_2^* = k_{21}$
$k_3^* = k_{31}$	$k_3^* = k_{31}$	$k_3^* = k_{31}$	k_{31}, k_{32}, k_{33}
	Vary k_1	Vary k_2	Vary k_3



Screening Design

- Screening design methodology: *One-factor-at-a-time approach*

- A model with k factors ($k \geq 2$)

- *One-factor-at-a-time approach*: Example

- Suppose that we have two factors A and B , for which we investigate the following two levels

The baseline levels of these factors are A^- and B^-

The proposed levels of these factors are A^+ and B^+

- According to the OFAT-method, we perform the following simulations

	Combination	Response
System 1	A^- and B^-	$R(A^-, B^-)$
System 2	A^+ and B^-	$R(A^+, B^-)$
System 3	A^- and B^+	$R(A^-, B^+)$

- The effect on the response of changing factor A from A^- to A^+ is computed as $R(A^+, B^-) - R(A^-, B^-)$

=> As a result of interaction effects, it is possible that the effect of changing factor A is different if factor B is at its B^+ level (i.e. if the factors interact)

=> In a 2^2 factorial design, we would also have simulated the combination A^+ and B^+ leading to a response $R(A^+, B^+)$

Screening Design

- Screening design methodology: *One-factor-at-a-time approach*
 - A model with k factors ($k \geq 2$) **(notes on the blackboard - not in your slides)**
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 - Suppose that we have two factors A and B , for which we investigate the following two levels

The baseline levels of these factors are A^- and B^-

The proposed levels of these factors are A^+ and B^+

Input values	Experiment 1	Experiment 2
$A^* = A^-$	A^-, A^+	$A^* = A^-$
$B^* = B^-$	$B^* = B^-$	B^-, B^+
	Vary A	Vary B

Screening Design

- Screening design methodology: *One-factor-at-a-time approach*

- A model with k factors ($k \geq 2$)

- *One-factor-at-a-time approach*: Example

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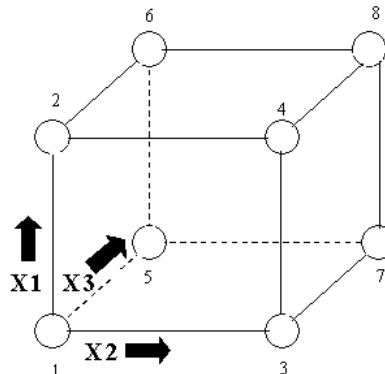
=> In a 2^2 factorial design, we would also have simulated the combination A^+ and B^+ leading to a response $R(A^+, B^+)$

Screening Design

- Screening design methodology: 2^k factorial design

- A model with k factors ($k \geq 2$)

- Determine factor impact and measure interactions
 - General model: $J(\alpha) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$ (number of factors $k = 2$)
 - 2^k factorial design
 - Select two levels for each of the k factors leading to 2^k possible factor-level combinations (called design points)
 - ‘−’ the lower numerical value (for quantitative factors)
 - ‘+’ the upper numerical value (for quantitative factors)
 - Run a number of simulations for each design point
 - Example: Design matrix - $k = 3$

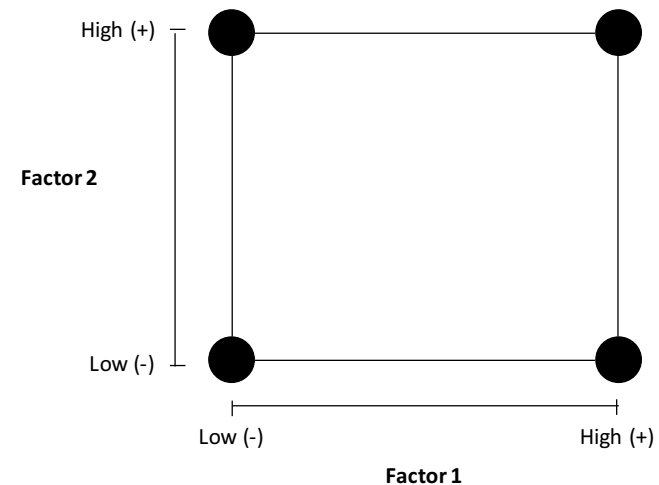


Factor combination (design point)	Factor 1	Factor 2	Factor 3	Response
1	−	−	−	R_1
2	+	−	−	R_2
3	−	+	−	R_3
4	+	+	−	R_4
5	−	−	+	R_5
6	+	−	+	R_6
7	−	+	+	R_7
8	+	+	+	R_8

Screening Design

- Screening design methodology: 2^k factorial design
 - A model with k factors ($k \geq 2$) **(notes on the blackboard - not in your slides)**
 - 2^2 factorial design
 - Select two levels for each of the 2 factors leading to 2^2 possible factor-level combinations (called design points)
 - ‘-’ the lower numerical value (for quantitative factors)
 - ‘+’ the upper numerical value (for quantitative factors)

Factor combination (design point)	Factor 1	Factor 2	Response
1	-	-	R_1
2	+	-	R_2
3	-	+	R_3
4	+	+	R_4



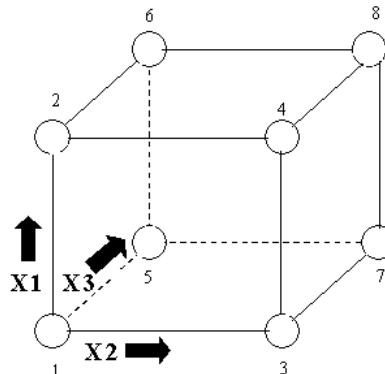
General model: $J(\alpha) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$ (number of factors $k = 2$)

Screening Design

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Factor combination (design point)	Factor 1	Factor 2	Factor 3	Response
1	−	−	−	R_1
2	+	−	−	R_2
3	−	+	−	R_3
4	+	+	−	R_4
5	−	−	+	R_5
6	+	−	+	R_6
7	−	+	+	R_7
8	+	+	+	R_8

Screening Design

- Screening design methodology: 2^k factorial design

➤ A model with k factors ($k \geq 2$): **Main effects**

- Main effect e_j of factor j is the average change in the response due to moving factor j from its '−' level to its '+' level while holding all other factors fixed,

i.e. the difference between the average response when factor j is at its '+' and the average response when it is at its '−' level

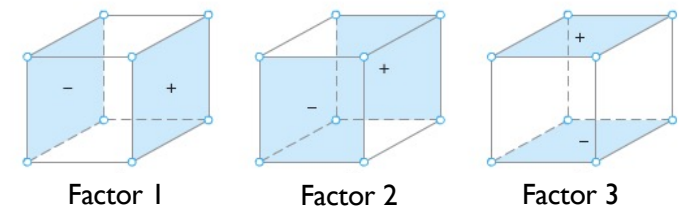
$$e_j = \frac{\sum_{l=1}^{2^k} a_{jl} R_l}{2^{k-1}} \quad \text{with } a_{jl} = 1 \text{ ('+' level) and } a_{jl} = -1 \text{ ('−' level)}$$

- Example: Design matrix - $k = 3$

$$e_1 = \frac{(R_2 - R_1) + (R_4 - R_3) + (R_6 - R_5) + (R_8 - R_7)}{4}$$

$$e_2 = \frac{(R_3 - R_1) + (R_4 - R_2) + (R_7 - R_5) + (R_8 - R_6)}{4}$$

$$e_3 = \frac{(R_5 - R_1) + (R_6 - R_2) + (R_7 - R_3) + (R_8 - R_4)}{4}$$



Factor combination				
(design point)	Factor 1	Factor 2	Factor 3	Response
1	−	−	−	R_1
2	+	−	−	R_2
3	−	+	−	R_3
4	+	+	−	R_4
5	−	−	+	R_5
6	+	−	+	R_6
7	−	+	+	R_7
8	+	+	+	R_8

Screening Design

- Screening design methodology: 2^k factorial design
 - A model with k factors ($k \geq 2$): **Main effects** (notes on the blackboard – not in your slides)
 - 2^2 factorial design
 - Main effect e_j of factor j is the average change in the response due to moving factor j from its '–' level to its '+' level while holding all other factors fixed,

$$e_j = \frac{\sum_{l=1}^{2^k} a_{jl} R_l}{2^{k-1}} \quad \text{with } a_{jl} = 1 \text{ ('+' level) and } a_{jl} = -1 \text{ ('-' level)}$$

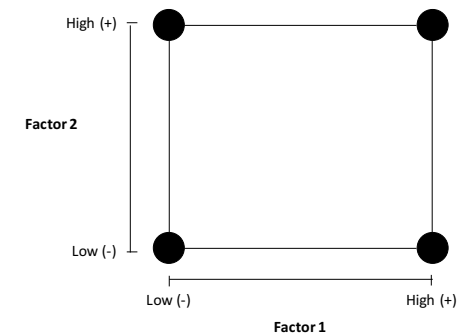
Average change when factor 2 is at its '-' level
Average change when factor 2 is at its '+' level

$$e_1 = \frac{(R_2 - R_1)}{2} + \frac{(R_4 - R_3)}{2} = \frac{-R_1 + R_2 - R_3 + R_4}{2}$$

Average change when factor 1 is at its '-' level
Average change when factor 1 is at its '+' level

$$e_2 = \frac{(R_3 - R_1)}{2} + \frac{(R_4 - R_2)}{2} = \frac{-R_1 - R_2 + R_3 + R_4}{2}$$

Factor combination (design point)	Factor 1	Factor 2	Response
1	-	-	R_1
2	+	-	R_2
3	-	+	R_3
4	+	+	R_4



Screening Design

- Screening design methodology: 2^k factorial design

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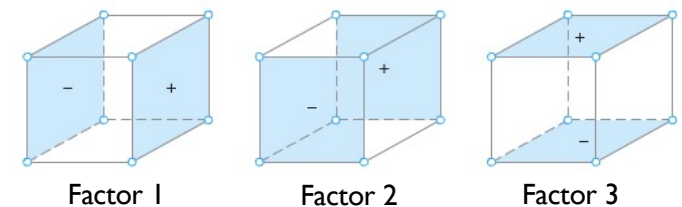
$$e_j = \frac{\sum_{l=1}^{2^k} a_{jl} R_l}{2^{k-1}} \quad \text{with } a_{jl} = 1 \text{ ('+' level) and } a_{jl} = -1 \text{ ('−' level)}$$

- Example: Design matrix - k = 3

$$e_1 = \frac{(R_2 - R_1) + (R_4 - R_3) + (R_6 - R_5) + (R_8 - R_7)}{4}$$

$$e_2 = \frac{(R_3 - R_1) + (R_4 - R_2) + (R_7 - R_5) + (R_8 - R_6)}{4}$$

$$e_3 = \frac{(R_5 - R_1) + (R_6 - R_2) + (R_7 - R_3) + (R_8 - R_4)}{4}$$



Factor combination				
(design point)	Factor 1	Factor 2	Factor 3	Response
1	-	-	-	R_1
2	+	-	-	R_2
3	-	+	-	R_3
4	+	+	-	R_4
5	-	-	+	R_5
6	+	-	+	R_6
7	-	+	+	R_7
8	+	+	+	R_8

Screening Design

- Screening design methodology: 2^k factorial design

➤ A model with k factors ($k \geq 2$): **Interaction effects**

- Two-factor (or two-way) interaction effect $e_{j_1 j_2}$ of factors j_1 and j_2 is the *one-half* difference between the average effect of factor j_1 when factor j_2 is at its '+' level and the average effect of factor j_1 when factor j_2 is at its '-' level

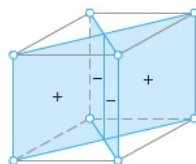
i.e. $j_1 \times j_2$ interaction $e_{j_1 j_2} = \frac{\sum_{l=1}^{2^k} a_{j_1 l} a_{j_2 l} R_l}{2^{k-1}}$ with $a_{jl} = 1$ ('+' level) and $a_{jl} = -1$ ('-' level)

- Example: Design matrix - $k = 3$

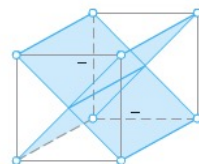
$$e_{12} = e_{21} = \frac{1}{2} \left[\frac{(R_4 - R_3) + (R_8 - R_7)}{2} - \frac{(R_2 - R_1) + (R_6 - R_5)}{2} \right]$$

$$e_{13} = e_{31} = \frac{R_1 - R_2 + R_3 - R_4 - R_5 + R_6 - R_7 + R_8}{4} = \frac{1}{2} \left[\frac{(R_6 - R_5) + (R_8 - R_7)}{2} - \frac{(R_2 - R_1) + (R_4 - R_3)}{2} \right]$$

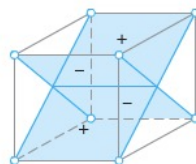
$$e_{23} = e_{32} = \frac{1}{2} \left[\frac{(R_7 - R_5) + (R_8 - R_6)}{2} - \frac{(R_3 - R_1) + (R_4 - R_2)}{2} \right]$$



Factor 1 x
Factor 3



Factor 1 x
Factor 2



Factor 2 x
Factor 3

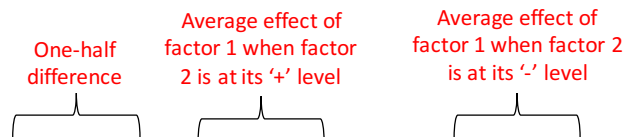
Factor combination (design point)	Factor 1	Factor 2	Factor 3	Response
1	-	-	-	R_1
2	+	-	-	R_2
3	-	+	-	R_3
4	+	+	-	R_4
5	-	-	+	R_5
6	+	-	+	R_6
7	-	+	+	R_7
8	+	+	+	R_8

Screening Design

- Screening design methodology: 2^k factorial design
 - A model with k factors ($k \geq 2$): **Interaction effects** (notes on the blackboard – not in your slides)
 - 2^2 factorial design
 - Two-factor (or two-way) interaction effect $e_{j_1 j_2}$ of factors j_1 and j_2 is the *one-half* difference between the average effect of factor j_1 when factor j_2 is at its '+' level and the average effect of factor j_1 when factor j_2 is at its '-' level

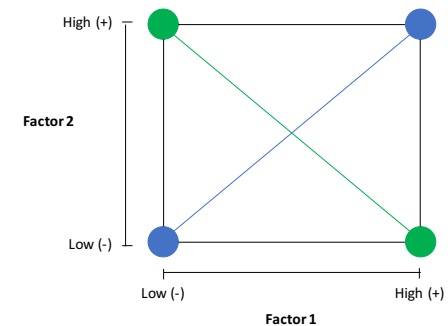
i.e. $j_1 \times j_2$ interaction $e_{j_1 j_2} = \frac{\sum_{l=1}^{2^k} a_{j_1 l} a_{j_2 l} R_l}{2^{k-1}}$

with $a_{jl} = 1$ ('+' level) and $a_{jl} = -1$ ('-' level)



- $e_{12} = e_{21} = \frac{1}{2} [(R_4 - R_3) - (R_2 - R_1)] = \frac{R_1 - R_2 - R_3 + R_4}{2}$

Factor combination (design point)	Factor 1 x		Response
	Factor 1	Factor 2	
1	-	-	R_1
2	+	-	R_2
3	-	+	R_3
4	+	+	R_4



Screening Design

- Screening design methodology: 2^k factorial design

➤ A model with k factors ($k \geq 2$): **Interaction effects**

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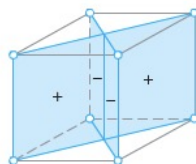
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- Example: Design matrix - $k = 3$

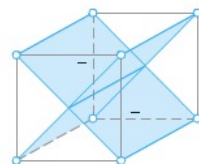
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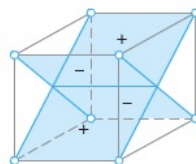
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Factor 1 x
Factor 3



Factor 1 x
Factor 2



Factor 2 x
Factor 3

Factor combination (design point)	Factor 1	Factor 2	Factor 3	Response
1	-	-	-	R_1
2	+	-	-	R_2
3	-	+	-	R_3
4	+	+	-	R_4
5	-	-	+	R_5
6	+	-	+	R_6
7	-	+	+	R_7
8	+	+	+	R_8

Screening Design

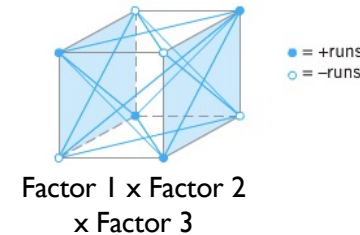
- Screening design methodology: 2^k factorial design

➤ A model with k factors ($k \geq 2$): **Interaction effects**

- Three-factor interaction effect $e_{j_1 j_2 j_3} = \frac{\sum_{l=1}^{2^k} a_{j_1 l} a_{j_2 l} a_{j_3 l} R_l}{2^{k-1}}$ with $a_{jl} = 1$ ('+' level) and $a_{jl} = -1$ ('-' level)

- Example: Design matrix - $k = 3$

$$e_{123} = e_{321} = \frac{-R_1 + R_2 + R_3 - R_4 + R_5 - R_6 - R_7 + R_8}{4} = e_{321} = e_{312} = \text{etc}$$



- General model: $J(\alpha) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_{12} x_1 x_2 + \beta_{13} x_1 x_3 + \beta_{23} x_2 x_3 + \beta_{123} x_1 x_2 x_3$
(number of factors $k = 3$)
- If two- or higher factor interactions are present, the main effect of each involved factor cannot be interpreted as simply the effect in general of moving that factor from its '-' level to its '+' level since the magnitude and the sign of the change in the response depend on the level of at least one other factor.

Screening Design

- Screening design methodology: 2^k factorial design

- Example: An Inventory Problem

- Experimental factors: Ordering policy

s the reorder point

d the reorder amount ($d = S - s$)

=> How do both factors impact the average total expected cost? Are there any interaction effects?

- Coding chart

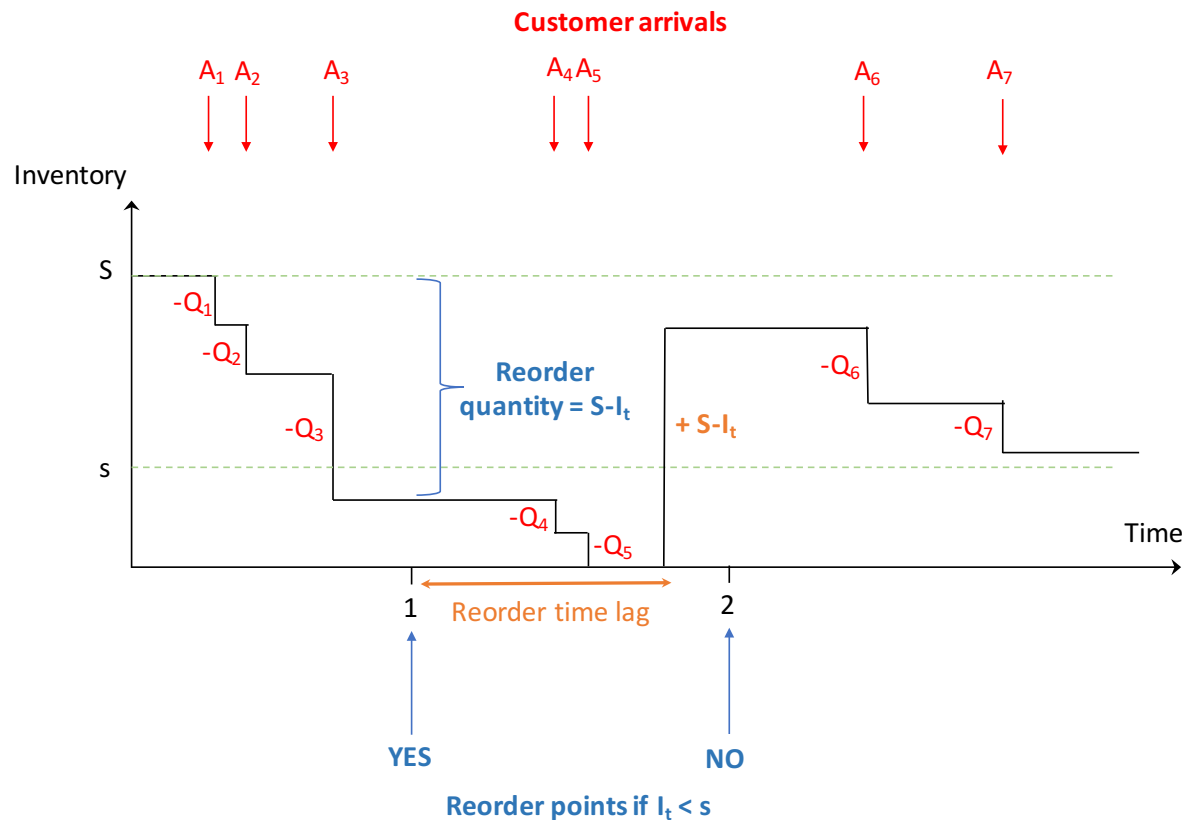
Factor	-	+
s	20	60
d	10	50

- 2^2 design matrix and simulation results (= the average cost per month from a single 120-month replication)

Factor combination				
(design point)	s	d	s x d	Response
1	-	-	+	144.16
2	+	-	-	144.50
3	-	+	-	119.99
4	+	+	+	147.00

Comparison between design alternatives

- (on the blackboard – not in your slides)
 - Comparison with a standard design – Example: An Inventory Problem



Screening Design

- Screening design methodology: 2^k factorial design

- Example: An Inventory Problem

- Experimental factors: Ordering policy

s the reorder point

d the reorder amount ($d = S - s$)

=> How do both factors impact the average total expected cost? Are there any interaction effects?

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3	-	+	-	119.99
4	+	+	+	147.00

Screening Design

- Screening design methodology: 2^k factorial design

- Example: An Inventory Problem

- Goal: Estimate regression model

$$J(\alpha) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 \text{ (number of factors } k = 2)$$

- Main effects

$$e_s = \frac{(R_2 - R_1) + (R_4 - R_3)}{2} = \frac{-144.16 + 114.50 - 119.99 + 147.00}{2} = 13.68 \quad \Rightarrow s \nearrow \text{cost} \nearrow$$

$$e_d = \frac{(R_3 - R_1) + (R_4 - R_2)}{2} = \frac{-144.16 - 114.50 + 119.99 + 147.00}{2} = -10.84 \quad \Rightarrow d \nearrow \text{cost} \searrow$$

- Interaction effects

$$e_{sd} = \frac{1}{2} [(R_4 - R_3) - (R_2 - R_1)] = \frac{144.16 - 114.50 - 119.99 + 147.00}{2} = 13.34 \quad \Rightarrow \text{setting } s \text{ and } d \text{ at opposite levels, cost } \searrow$$

Screening Design

- Screening design methodology: 2^k factorial design
 - A model with k factors ($k \geq 2$): **Statistical significance of effects**
 - The output performance is a random variable
 - In order to determine the effect is statistically significant and not to be explained by sampling fluctuation, we must perform an **analysis of variance**
 - Replicate the whole design n times to obtain n independent and identically distributed values of each effect

Calculate mean $\bar{e}_j(n) = \frac{\sum_{i=1}^n e_j^i}{n}$ with e_j^i the main effect for factor j for the i^{th} simulation run

Calculate variance $S_j^2(n) = \frac{\sum_{i=1}^n (e_j^i - \bar{e}_j(n))^2}{n-1}$

Calculate 100(1- α)% confidence interval: $[\bar{e}_j(n) - \frac{t_{n-1, 1-\frac{\alpha}{2}} S_j(n)}{\sqrt{n}}, \bar{e}_j(n) + \frac{t_{n-1, 1-\frac{\alpha}{2}} S_j(n)}{\sqrt{n}}]$

=> If the confidence interval for $E(e_j)$ does not contain 0, we can conclude that the effect is statistically significant, otherwise not

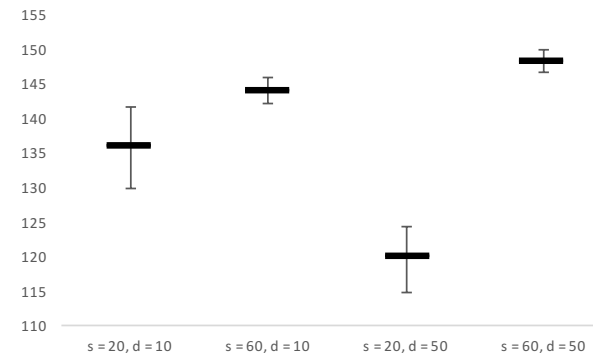
Screening Design

- Screening design methodology: 2^k factorial design

➤ Example: An Inventory Problem

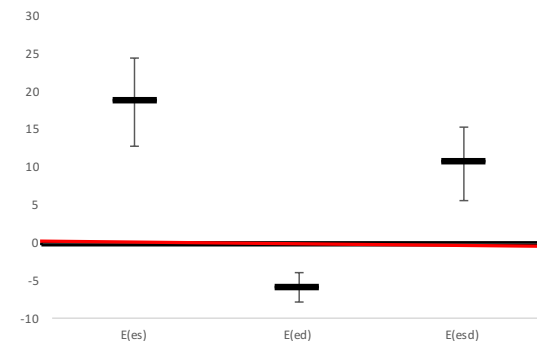
- Results: $n = 10$ independent simulation runs of the entire 2^2 factorial design

Design point	Sample mean	Sample variance
$s = 20, d = 10$	135.71	22.24
$s = 60, d = 10$	143.94	2.26
$s = 20, d = 50$	119.45	15.07
$s = 60, d = 50$	148.17	1.60



- The 96.67% confidence intervals (resulting in an overall confidence level of at least 90%)

Expected effect	96.67% confidence
	Interval
$E(e_s)$	18.47 ± 2.58
$E(e_d)$	-6.02 ± 2.47
$E(e_{sd})$	10.25 ± 2.88



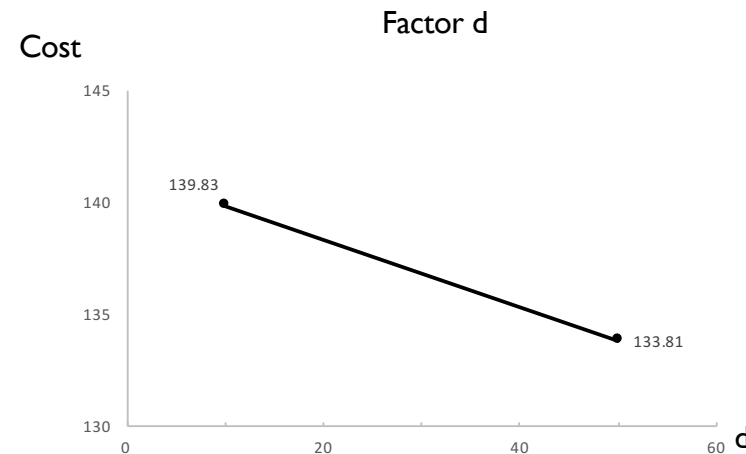
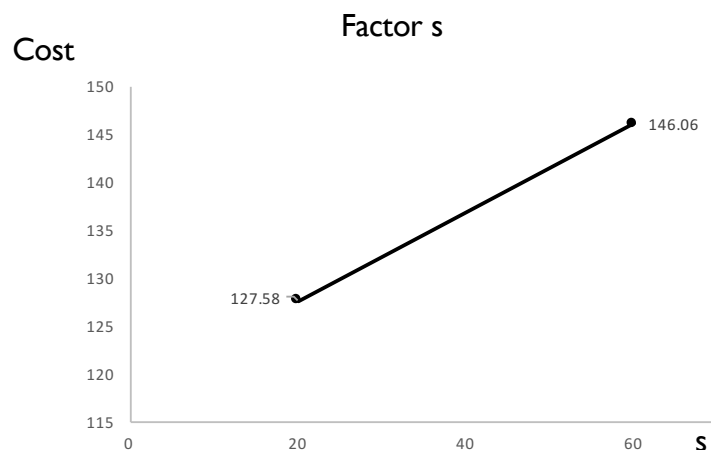
The main effects and interaction effect are significant!

Screening Design

- Screening design methodology: 2^k factorial design

- Example: An Inventory Problem

- Main-effect plots (results are averaged over the two levels of the other factor)



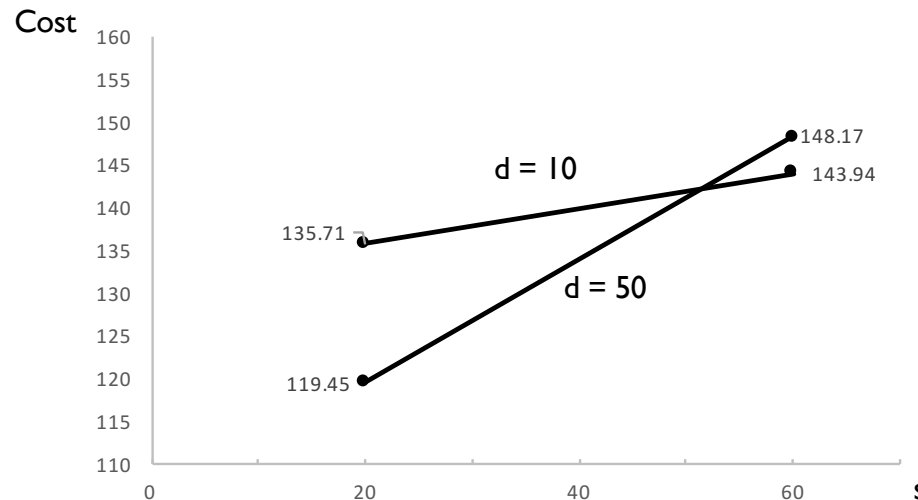
=> However, since there is a significant interaction between s and d, these main effects are of limited value

Screening Design

- Screening design methodology: 2^k factorial design

- Example: An Inventory Problem

- Interaction plot for s and d



⇒ The presence of a significant interaction is indicated by the non-parallel lines

⇒ When lines are parallel there is no interaction effect

Screening Design

- Screening design methodology: 2^k factorial design
 - Some applications in literature
 - Hood, S.J., and Welch, P.D., 1992, Experimental Design Issues in Simulation with Examples from Semiconductor Manufacturing, Proceedings 1992 Winter Simulation Conference, Washington D.C., 255-263
 - Kang, K., Doerr, K.H., Sanchez, S.M., 2006, A design of experiments approach to readiness risk analysis, Proceedings 2006 Winter Simulation Conference, Monterey, California, 1332-1339
 - Kumar, S., and Nottestad, D.A., 2006, Capacity Design: An Application Using Discrete-Event Simulation and Designed Experiments, IIE Transactions, 38, 729-736
 - Porcaro, D.M., 1996, Simulation Modelling and DOE, Industrial Engineering Solutions, 24-30.
 - Posadas, S., and Paulo, E.P., 2003, Stochastic Simulation of a Commander's Decision Cycle, Military Operations Research, 8, 21-43.

Screening Design

- Screening design methodology: 2^{k-p} fractional factorial design
 - Disadvantage of 2^k factorial design
 - A full 2^k factorial design becomes quickly unmanageable
 - E.g. $k = 11$: $2^{11} = 2048$ design points $\times$ 5 replications leads to 10240 replications
 - Principle 2^{k-p} fractional factorial design
 - Screen out an important subset of k factors of interest
 - Fractional factorial design provide a way to get a good estimate of the main effects and the two-way interactions at a fraction of the CPU required by a full 2^k factorial design
 - Method
 - Select a subset of size 2^{k-p} of all the 2^k possible design points
 - Run a simulation for this fraction ($1/2^p$) of the possible combinations (if $p = 1$: half fraction; $p = 2$; quarter fraction; etc.)
 - The larger p , the smaller the required CPU time, but the less information from the experiment

Screening Design

- Screening design methodology: 2^{k-p} fractional factorial design

➤ Principle 2^{k-p} fractional factorial design

- Which 2^{k-p} design points should we select out of the possible 2^k combinations?

- Based on the idea of confounding

We may find exactly the same algebraic expression for several different effects

E.g. in a 2^{4-1} half fraction, the formulas for the main effect e_4 and the three-way interaction effect e_{123} are identical

=> The main effect for factor 4 is *confounded* or *aliased* with the three-way interaction effect between factors 1, 2 and 3

=> e_4 and e_{123} are aliases of each other, i.e. $e_4 = e_{123}$

- Construct design matrix

Identify p and determine which 2^{k-p} rows to use of the full 2^k factorial design

Write out the the full 2^{k-p} factorial design matrix in factor 1, 2, ..., k-p

The remaining p columns are determined by multiplying the first k-p columns

E.g. Design point 1: Level Factor 4 = Factor 1 x Factor 2 x Factor 3

= '−' level x '−' level x '−' level = '−' level

Factor combination (design point)	Factor 1	Factor 2	Factor 3	Factor 4
1	−	−	−	−
2	+	−	−	+
3	−	+	−	+
4	+	+	−	−
5	−	−	+	+
6	+	−	+	−
7	−	+	+	−
8	+	+	+	+

Screening Design

- Screening design methodology: 2^{k-p} fractional factorial design

➤ Example 2^{k-p} fractional factorial design: A 2^{3-1} design

- Consider the two-level, full-factorial design for three factors, i.e. the 2^3 design

Factor combination (design point)	Factor 1	Factor 2	Factor 3	R_i
1	-	-	-	33
2	+	-	-	63
3	-	+	-	41
4	+	+	-	57
5	-	-	+	57
6	+	-	+	51
7	-	+	+	59
8	+	+	+	53

- Based upon the responses, we can compute the main effects e.g.

$$e_1 = \frac{(R_2 - R_1) + (R_4 - R_3) + (R_6 - R_5) + (R_8 - R_7)}{4} = 8.5$$

$$e_2 = \frac{-R_1 - R_2 + R_3 + R_4 - R_5 - R_6 + R_7 + R_8}{4} = 1.5$$

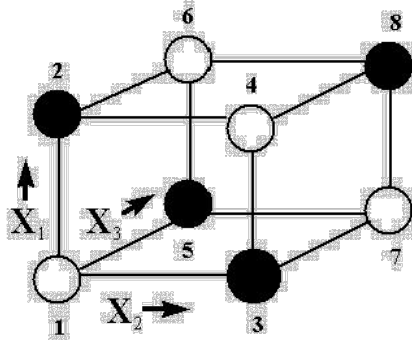
$$e_3 = \frac{-R_1 - R_2 - R_3 - R_4 + R_5 + R_6 + R_7 + R_8}{4} = 6.5$$

Screening Design

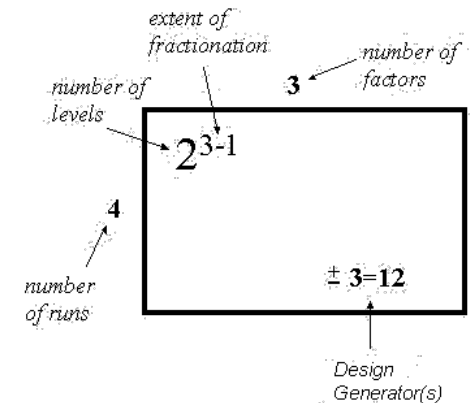
- Screening design methodology: 2^{k-p} fractional factorial design

➤ Example 2^{k-p} fractional factorial design: A 2^{3-1} design

- Suppose that we want to reduce the number of design points to investigate and select only the four shaded corners of the cube



Factor combination (design point)	Factor 1	Factor 2	Factor 3 = Factor 1 x Factor 2
1	-	-	+
2	+	-	-
3	-	+	-
4	+	+	+



- Based upon the responses, we can compute the main effects e.g.

$$e_1 = \frac{-R_1 + R_2 + R_3 - R_4}{2} = 8 \quad e_2 = \frac{-R_1 - R_2 + R_3 + R_4}{2} = 16 \quad e_3 = \frac{R_1 - R_2 - R_3 + R_4}{2} = 10$$

=> Main effect estimates from fractional factorial not as good as full factorial

- We also could have used the unshaded corners, i.e. Factor 3 = - Factor 1 x Factor 2

Screening Design

- Screening design methodology: 2^{k-p} fractional factorial design

➤ Example 2^{k-p} fractional factorial design: A 2^{8-3} design

- The fractional factorial design has 32 runs and eight factors
- How to construct the design
 - Write down the full factorial design in standard order for k-p factors (=8-3=5)

=> $2^5 = 32$ rows

- Add additional columns to to the design table with

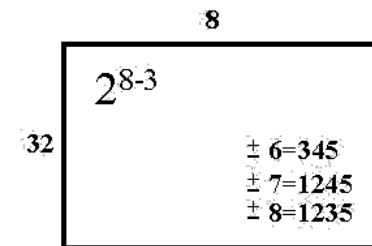
Factor 6 = Factor 3 x Factor 4 x Factor 5

Factor 7 = Factor 1 x Factor 2 x Factor 4 x Factor 5

Factor 8 = Factor 1 x Factor 2 x Factor 3 x Factor 5

=> The resultant design matrix gives the 32 trial runs for an 8-factor fractional factorial design

- Further reading/Relevant tables for constructing fractional factorial designs
 - Montgomery, D.C., 2013, Design and Analysis of Experiments, 8th Edition, John Wiley, New York
 - Law, A.M., 2015, Simulation Modelling and Analysis, 5th Edition, McGrawHill, New York



Screening Design

- Screening design methodology: 2^{k-p} fractional factorial design
 - Rules for constructing 2^{k-p} fractional factorial designs

Runs	Factors (k)						
	3	4	5	6	7	8	9
4	2^{3-1}_{III} 3 = ± 12						
8		2^{4-1}_{IV} 4 = ± 123	2^{5-2}_{III} 4 = ± 12 5 = ± 13	2^{6-3}_{III} 4 = ± 12 5 = ± 13 6 = ± 23	2^{7-4}_{III} 4 = ± 12 5 = ± 13 6 = ± 23 7 = ± 123		
16			2^{5-1}_V 5 = ± 1234	2^{6-2}_{IV} 5 = ± 123 6 = ± 234	2^{7-3}_{IV} 5 = ± 123 6 = ± 234 7 = ± 134	2^{8-4}_{IV} 5 = ± 123 6 = ± 234 7 = ± 134 8 = ± 124	2^{9-5}_{III} 5 = ± 123 6 = ± 234 7 = ± 134 8 = ± 124 9 = ± 1234
32				2^{6-1}_{VI} 6 = ± 12345	2^{7-2}_{IV} 6 = ± 1234 7 = ± 1245	2^{8-3}_{IV} 6 = ± 123 7 = ± 124 8 = ± 2345	2^{9-4}_{IV} 6 = ± 2345 7 = ± 1345 8 = ± 1245 9 = ± 1235
64					2^{7-1}_{VII} 7 = ± 123456	2^{8-2}_V 7 = ± 1234 8 = ± 1256	2^{9-3}_{IV} 7 = ± 1234 8 = ± 1356 9 = ± 3456
128							2^{9-2}_{VI} 8 = ± 13467 9 = ± 23567

Response Surface Design

- Response surface methodology

- Steps

1. The experimental strategy for exploring the space of the process or independent variables,
2. Empirical statistical modeling to develop an appropriate approximating relationship between the response and the input factors. Repeat steps 1 and 2 to find the region of the optimum
3. Optimization methods for finding the values of the process variables that produce desirable values of the response

- Methodology

- Build a metamodel or response surface that transforms a particular set of input factor values into the output response via mathematical/statistical techniques
 - Analytical metamodel (e.g. M/M/1 queue)
 - Empirical metamodel for most systems (based on statistical/regression analysis)
- Seek for the settings of the input factors that optimises the response via optimisation methods
 - Stochastic Approximation
 - Alternative Heuristic optimisation: Simulated Annealing

Response Surface Design

- Determine the optimal design α^* that optimises the output $J(\alpha)$

$$\alpha^* = \arg \min_{\alpha \in \mathcal{A}} J(\alpha)$$

$$\text{with } \alpha = (v_1, v_2, \dots, v_k)$$

$$\text{s.t. } l_i \leq v_i \leq u_i \quad \forall i \quad (\text{boundary values on characteristic } i)$$

$$\sum_i a_{pi} v_i \leq b_p \quad \forall p \quad (\text{more general constraints})$$

- For simple systems: Analytical model (closed-form)
- For complex model: Regression model – Predict response given factor inputs
 - Consider only the significant factors (determined via the Screening Design Methodology)
 - First order model with interaction (number of factors = 2)

$$J(\alpha) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$$

- Second order model with interaction (number of factors = 2)

$$J(\alpha) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2$$

=> Very flexible and good approximation of true response surface

=> Estimate β via least squares method

=> Use Central Composite Design and generate center points

Response Surface Design

- Determine the optimal design α^* that optimises the output $J(\alpha)$

$$\alpha^* = \arg \min_{\alpha \in \mathcal{A}} J(\alpha)$$

$$\text{with } \alpha = (v_1, v_2, \dots, v_k)$$

$$\text{s.t. } l_i \leq v_i \leq u_i \quad \forall i$$

(boundary values on characteristic i)

$$\sum_i a_{pi} v_i \leq b_p \quad \forall p$$

(more general constraints)

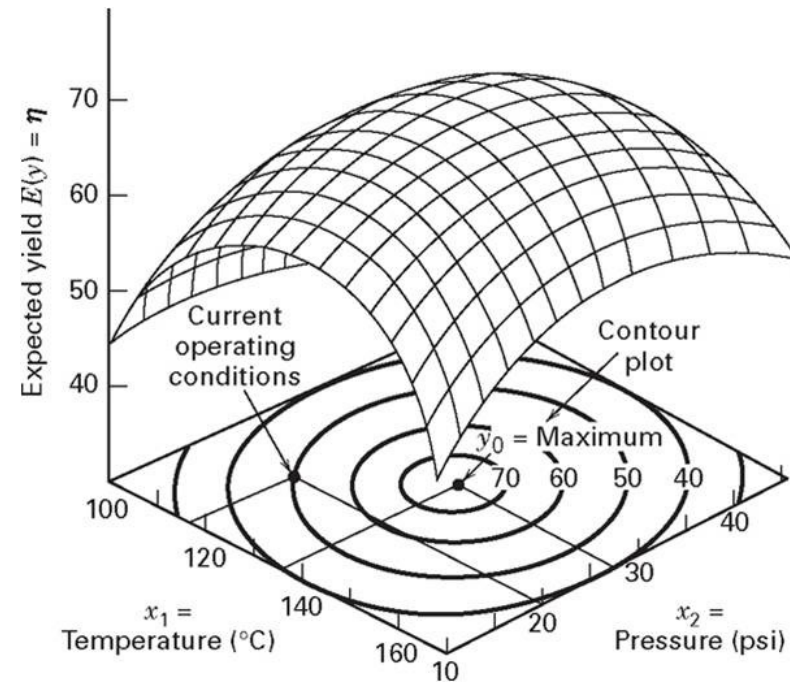
➤ Assumptions

- A single global minimum exists in $\mathcal{A}$
- $\mathcal{A}$ is either a continuous or discrete set

➤ Focus: Find the optimal design of a system

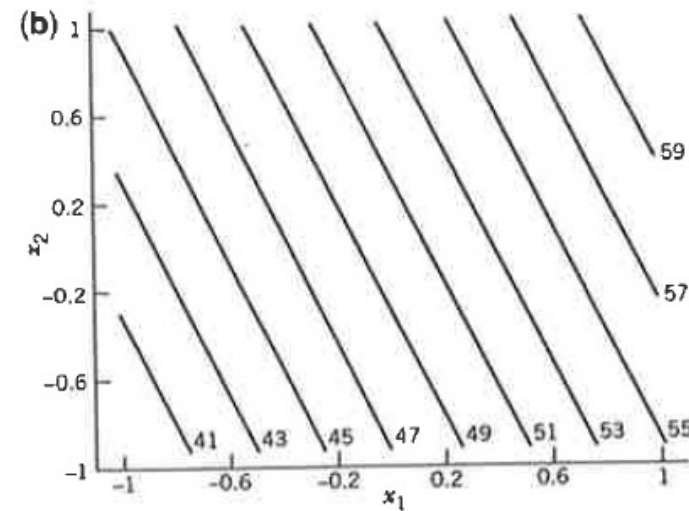
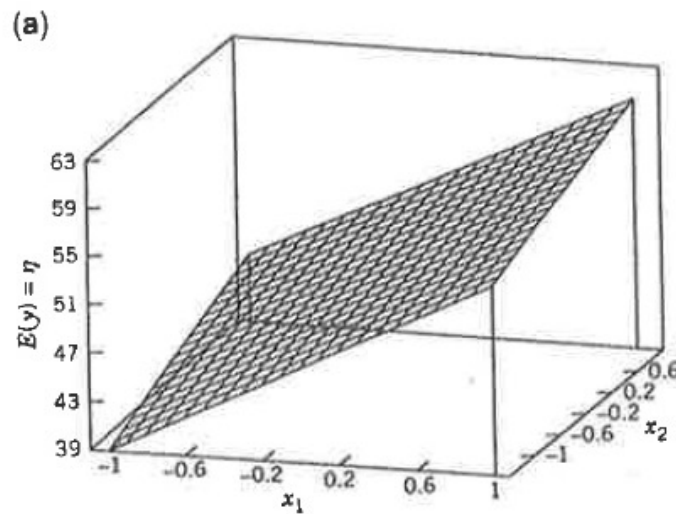
Response Surface Design

- Graphical representation
 - Response surface plot vs contour plot



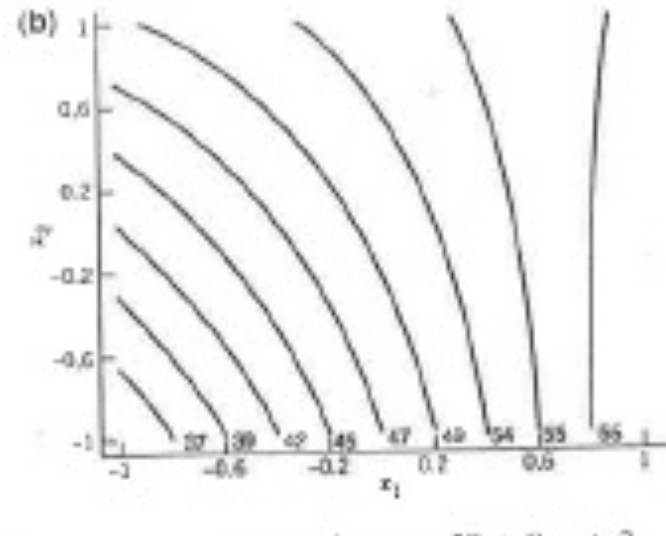
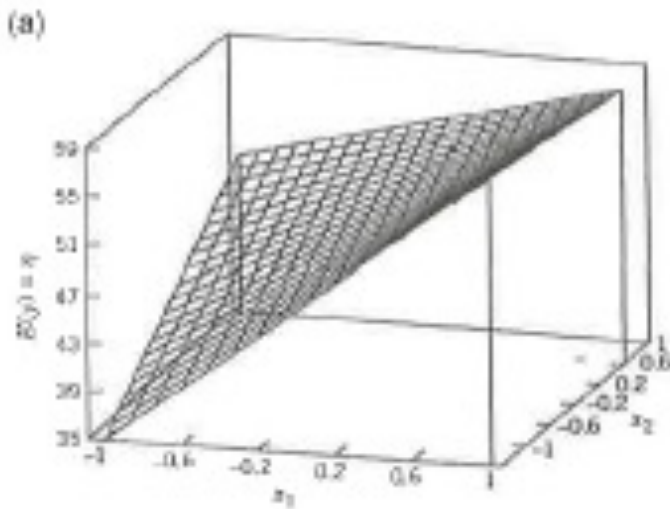
Response Surface Design

- Graphical representation
 - Response surface plot (a) and contour plot (b) for the first order model
$$J(\alpha) = 50 + 8x_1 + 3x_2$$



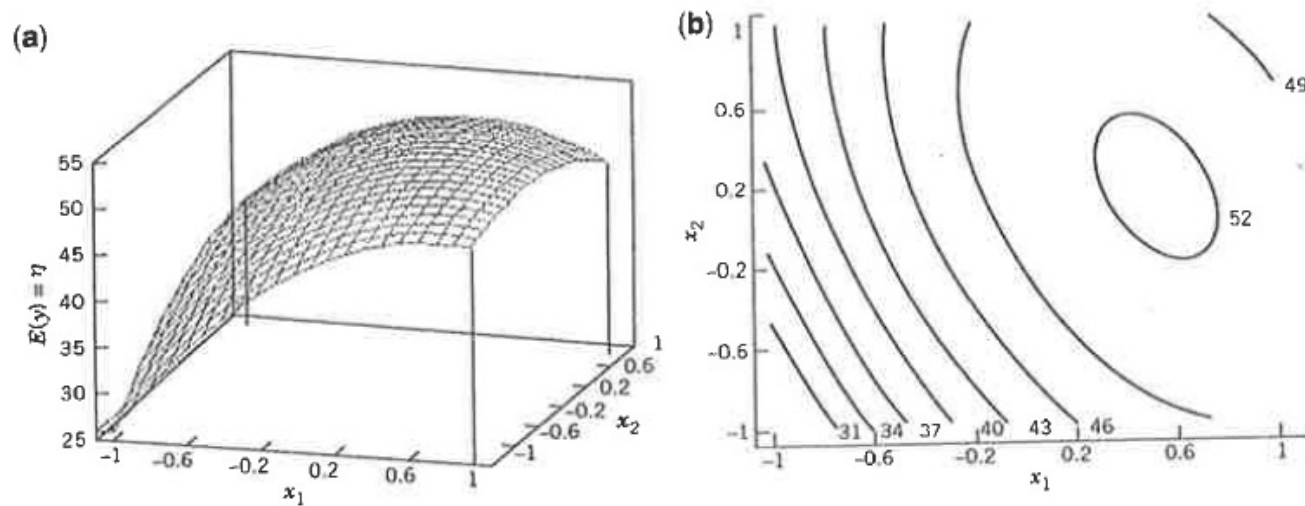
Response Surface Design

- Graphical representation
 - Response surface plot (a) and contour plot (b) for the first order model with interaction:
$$J(\alpha) = 50 + 8x_1 + 3x_2 - 4x_1x_2$$



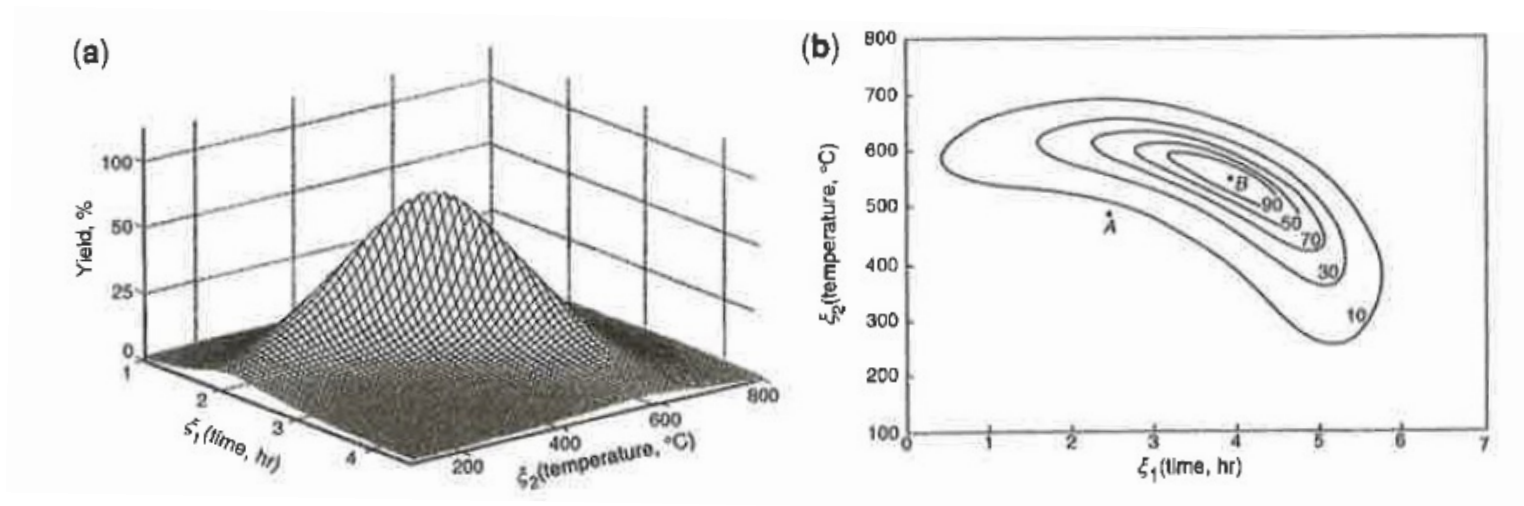
Response Surface Design

- Graphical representation
 - Response surface plot (a) and contour plot (b) for the second order model
$$J(\alpha) = 50 + 8x_1 + 3x_2 - 7x_1^2 - 3x_2^2 - 4x_1x_2$$



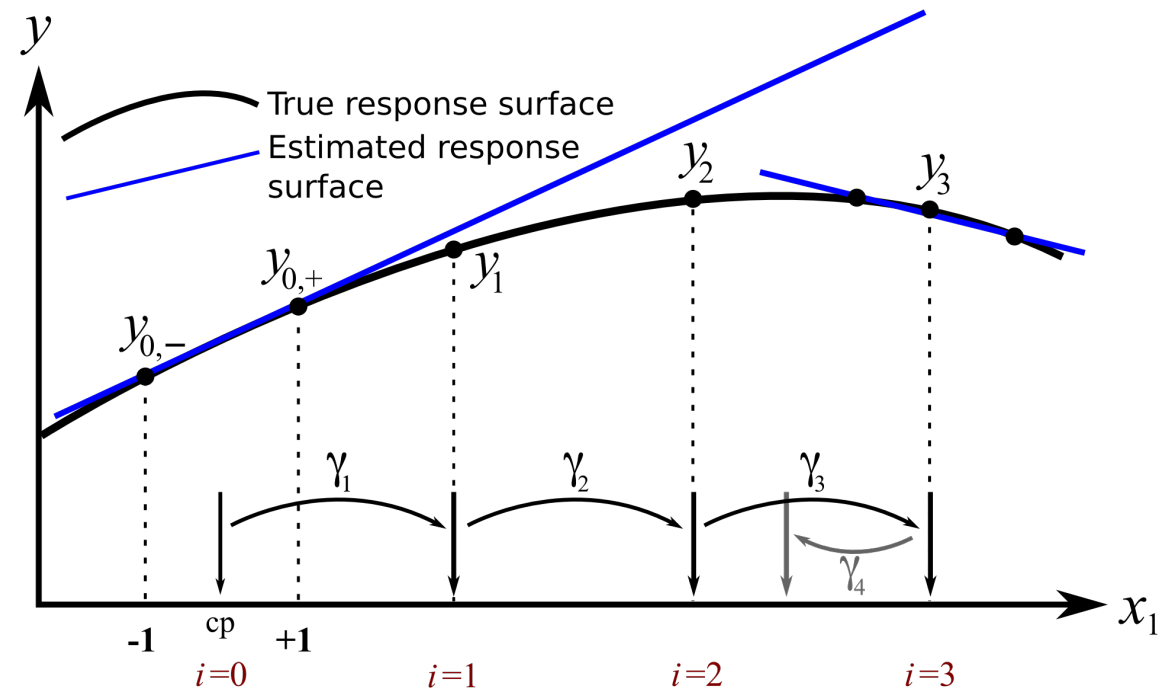
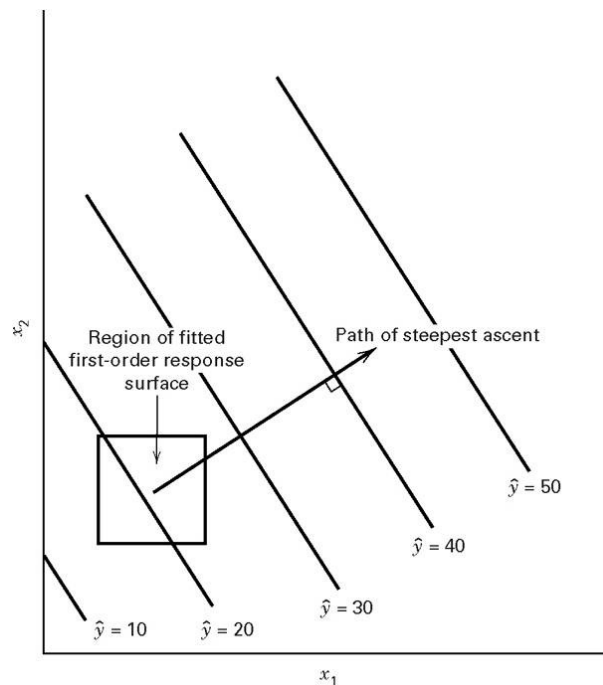
Response Surface Design

- Graphical representation
 - Response surface plot (a) and contour plot (b) for a higher-order model



Response Surface Design

- Response surface design methodology is a sequential procedure
- Interaction First-order and Second-order model



Steepest ascent

=> Moving sequentially from an initial guess to the region of the optimum

=> Need for first-order model

Stagnation

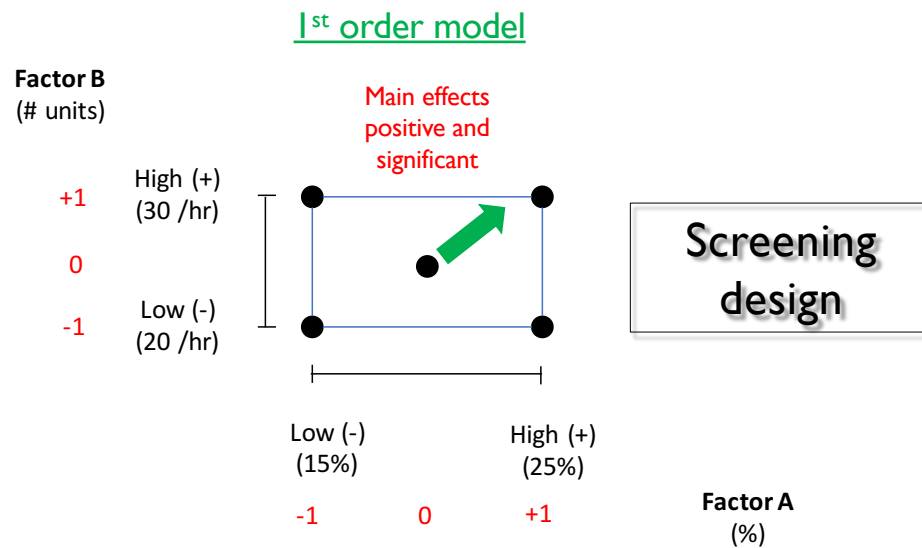
=> Curvature expected

=> Need for second-order model

Response Surface Design

- Response surface design methodology is a sequential procedure
(notes on the blackboard – not in your slides)

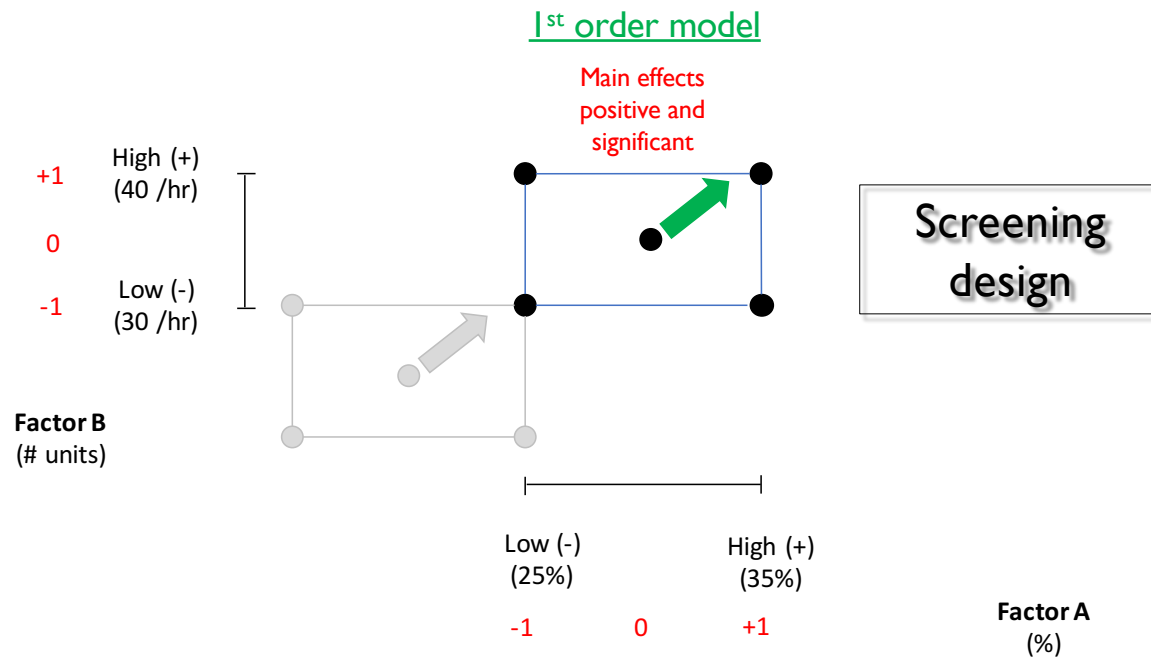
ITERATION I



Response Surface Design

- Response surface design methodology is a sequential procedure
(notes on the blackboard – not in your slides)

ITERATION 2



Response Surface Design

- Response surface design methodology is a sequential procedure
(notes on the blackboard – not in your slides)

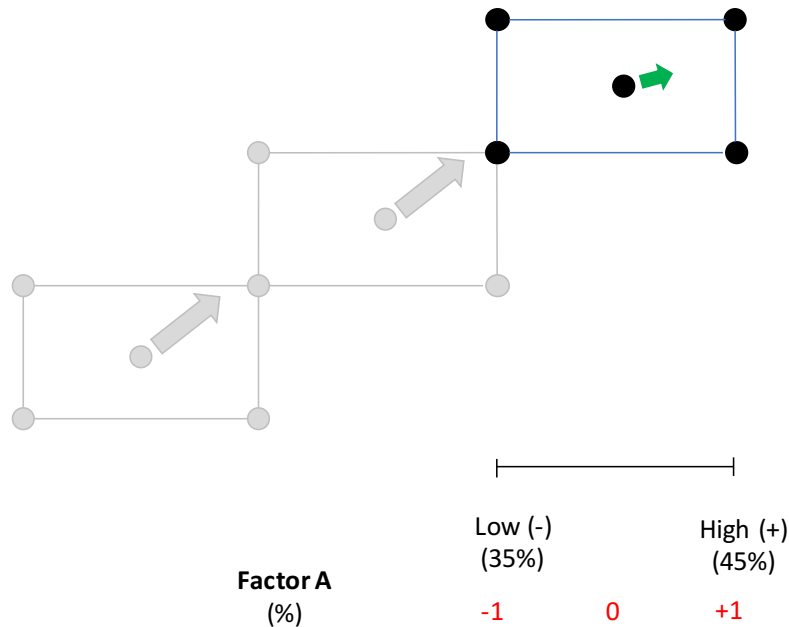
1st order model

Main effects
positive but
insignificant

ITERATION 3

+1 High (+)
(50 /hr)
0
-1 Low (-)
(40 /hr)

Factor B
(# units)

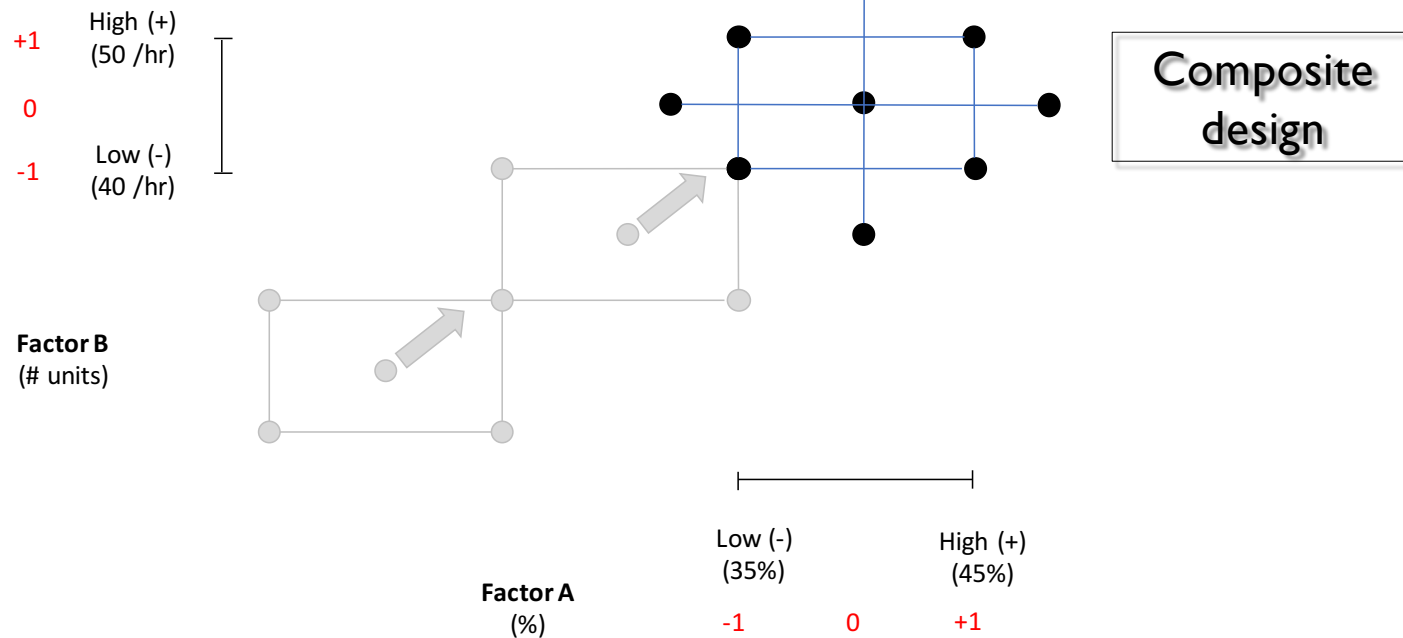


Screening
design

Response Surface Design

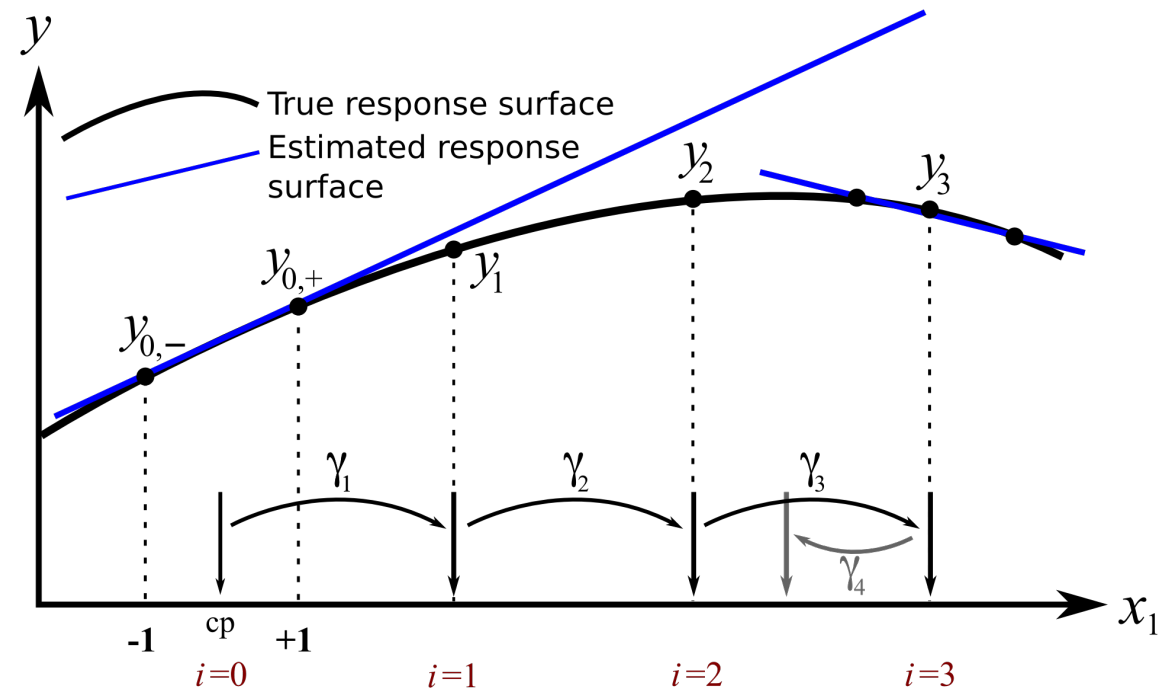
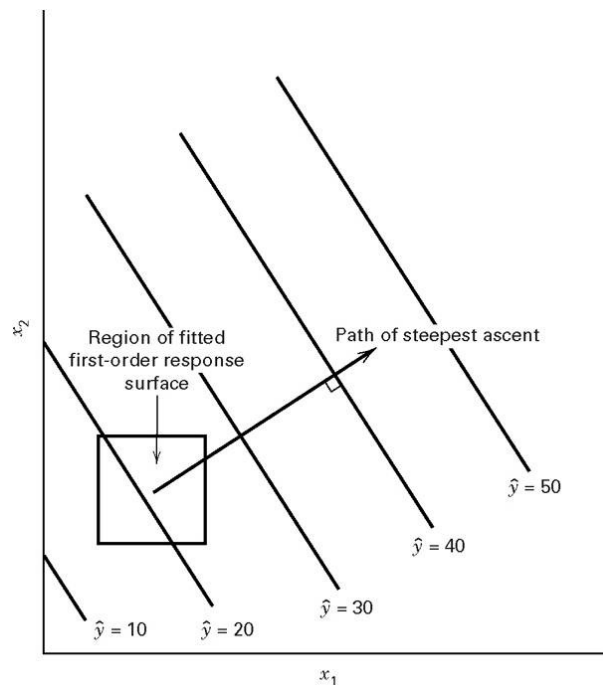
- Response surface design methodology is a sequential procedure
(notes on the blackboard – not in your slides)

ITERATION 4



Response Surface Design

- Response surface design methodology is a sequential procedure
- Interaction First-order and Second-order model



Steepest ascent

=> Moving sequentially from an initial guess to the region of the optimum

=> Need for first-order model

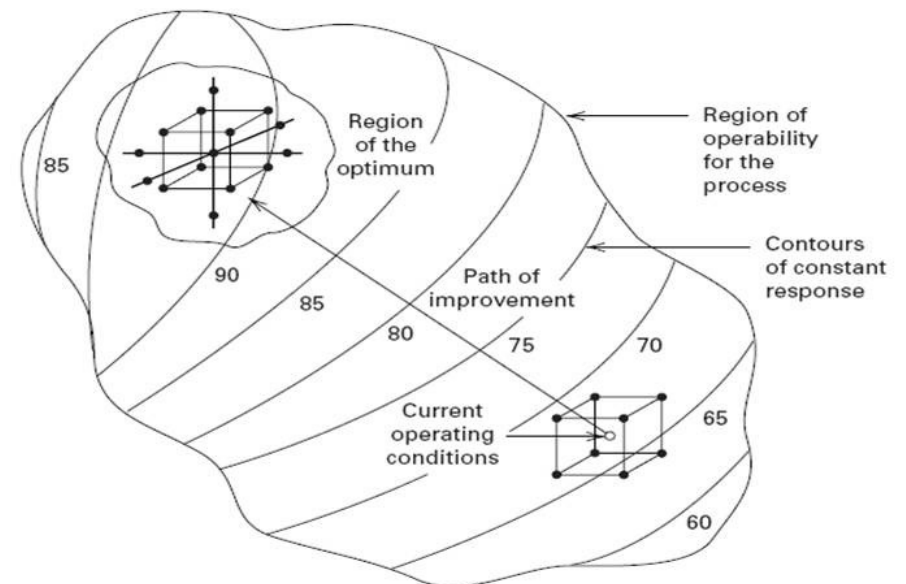
Stagnation

=> Curvature expected

=> Need for second-order model

Response Surface Design

- Response surface design methodology is a sequential procedure
 - Factor screening
 - Finding the region of the optimum
 - Modeling & Optimisation of the response



■ FIGURE 11.3 The sequential nature of RSM

Response Surface Design

- Model representation

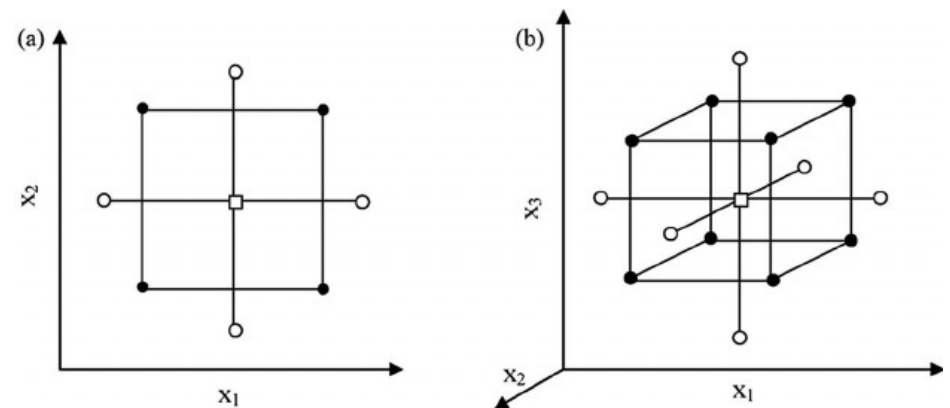
- Simulate different design points leading to different outcomes for the response variable(s)

- First-order model

- 2^k full factorial or fractional factorial design
 - 2^k full factorial or fractional factorial design with center points ($x_i = 0$)

- Second-order model

- Central composite design (a) two variables (b) three variables
 - Box-Behnken design
 - Other



- Estimate the regression coefficients via multiple linear/non-linear regression

- Only include those terms with a coefficient significantly different from 0

Response Surface Design

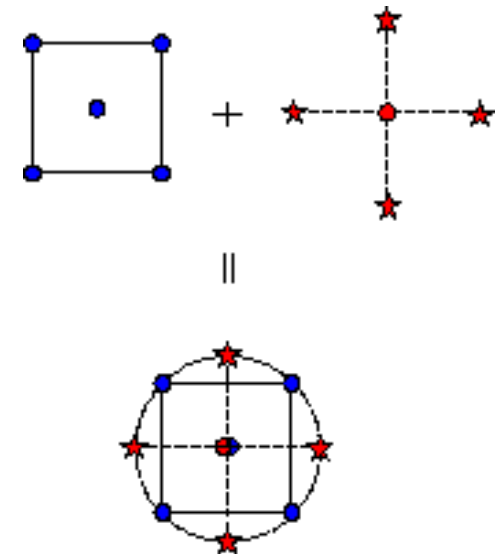
- Central composite model explained

- Composition

- Full or fractional factorial design with center points
 - Group of star points to allow the estimation of curvature

- Three sets of experimental runs

- A factorial design of certain factors, each having two levels ($\# = 2^k$)
 - A set of center points (multiple replications $\geq 2 \times k$)
 - A set of star or axial points ($\# = 2 \times k$), experimental runs identical to the center point except for one factor, which is set to an extreme value α , typically outside the range. The set of star points are at some value α or $-\alpha$ on each axis.

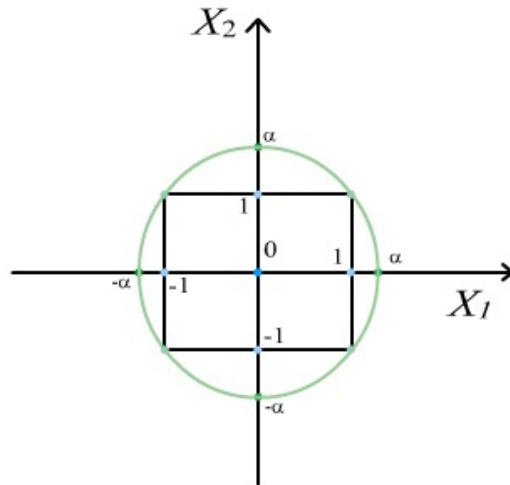


Response Surface Design

- Central composite model explained

- Choice of α

- If $\alpha = 1$, the star points are right on the boundary of the box leading to a 3^k design (special case: Face centered composite design)
- If $\alpha = \sqrt{k}$, the star points lie outside the box of the 2^k design, leading to a spherical design (common case: Circumscribed composite design) e.g. $\alpha = \sqrt{2}$ when $k = 2$

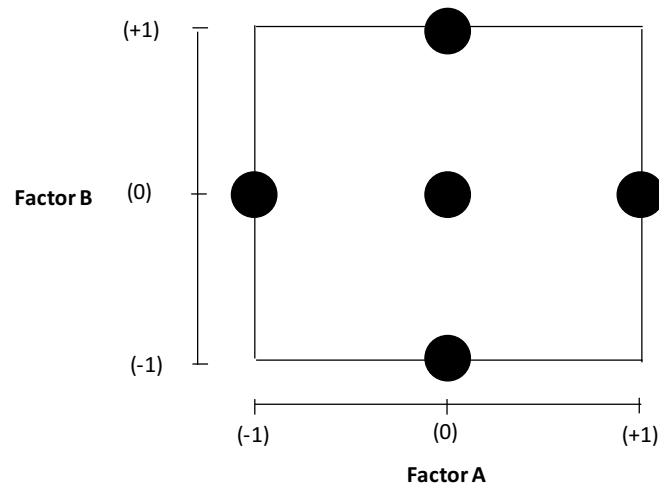


Design point	Factor 1	Factor 2	Response
1	-1	-1	R_1
2	1	-1	R_2
3	-1	1	R_3
4	1	1	R_4
5	-1.414	0	R_5
6	1.414	0	R_6
7	0	-1.414	R_7
8	0	1.414	R_8
9	0	0	R_9
10	0	0	R_{10}
11	0	0	R_{11}
12	0	0	R_{12}

Response Surface Design

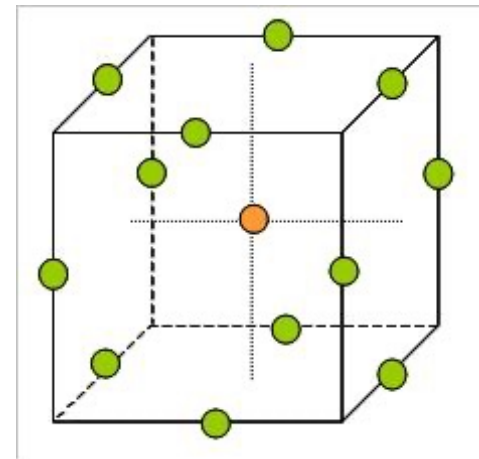
- Box Behnken model explained (notes on the blackboard not in your slides)

(a) Two-factor design



A	B	Design points
-1	0	Factorial point
+1	0	Factorial point
0	-1	Factorial point
0	+1	Factorial point
0	0	Centre points

(b) Three-factor design



A	B	C	Design points	Number of runs
± 1	± 1	0	Factorial points	4
± 1	0	± 1	Factorial points	4
0	± 1	± 1	Factorial points	4
0	0	0	Centre points	$1 \times 5 = 5$
Total				17

Optimal design of the system

- Stochastic Approximation

- Derivatives of the cost function

- Parameter sensitivity: Indicate how sensitive a cost $J(\alpha)$ is to design parameter α

- Assumptions: $\mathcal{A}$ is continuous and $J(\alpha)$ is differentiable and convex, i.e. $J''(\alpha) > 0, \forall \alpha \in \mathcal{A}$, leading to a unique optimal solution

- First derivative: $J'(\alpha) = \frac{dJ(\alpha)}{d\alpha}$ (Estimator: $\bar{J}'(\alpha)$)

- Second derivative: $J''(\alpha) = \frac{d^2J(\alpha)}{d\alpha^2}$ (Estimator: $\bar{J}''(\alpha)$)

Optimal design of the system

- Stochastic Approximation

- Derivatives of the cost function

- Used in optimisation procedures

- Derivatives can be used to guess in what direction to search for α^*
 - Using the estimate of the first derivate $\bar{J}'(\alpha) = 0$ to find α^* is inaccurate/very slow
 - Use Newton's algorithm (Non-linear unconstrained optimisation): Iterative algorithm that generates a sequence of trial solutions ($\alpha_i \rightarrow \alpha_{i+1}$) as follows

$$\alpha_{i+1} = \alpha_i - \frac{J'(\alpha_i)}{J''(\alpha_i)} \quad (\text{iteration } i)$$

The algorithm converges towards the optimal solution α^*

Instead of evaluating $\bar{J}''(\alpha_i)$, we make use of a fixed sequence $k_i \rightarrow 0$ (to avoid oscillation around α^*)

$$\alpha_{i+1} = \alpha_i - k_i J'(\alpha_i)$$

Optimal design of the system

- Stochastic Approximation

- Derivatives of the cost function

- Estimation of first derivative $J'(\alpha)$

- Use arbitrary small increment $d\alpha$
 - Generate $Y_k(\alpha)$ and $Y_k(\alpha + d\alpha)$ with common random numbers

$$Y_k(\alpha) \quad \rightarrow \quad \bar{J}(\alpha) = \frac{1}{K} \sum_{k=1}^K h(\alpha, Y_k(\alpha))$$

$$Y_k(\alpha + d\alpha) \quad \rightarrow \quad \bar{J}(\alpha + d\alpha) = \frac{1}{K} \sum_{k=1}^K h(\alpha + d\alpha, Y_k(\alpha + d\alpha))$$

- The estimate of the first derivative

$$J'(\alpha) = \frac{J(\alpha + d\alpha) - J(\alpha)}{d\alpha} = \frac{\sum_{k=1}^K (h(\alpha + d\alpha, Y_k(\alpha + d\alpha)) - h(\alpha, Y_k(\alpha)))}{K d\alpha} \quad (\text{unbiased estimator})$$

Optimal design of the system

- Example 1: Determine the optimal service speed in a queueing system

- Problem setting

- Consider a queueing system with a single server and a single stream of arriving tasks (customers). The optimal design problem must determine the optimal service speed α^* in order to minimise cost

- Input

- Arrival process: Poisson distributed with parameter λ ($F_A(\alpha, a)$)
- Service process: Exponentially distributed with mean μ ($F_S(\alpha, s)$)
- Task k experiences a delay D_k

- Cost function

- Design parameter α is the server speed: If α increases, ...
 - The server rate μ increases
 - The operational/implementation cost $C(\alpha)$ increases
 - The average delay $E[D(\alpha)]$ probably decreases
- Cost function: $J(\alpha) = C(\alpha) + cE[D(\alpha)]$

Optimal design of the system

- Example 1: Determine the optimal service speed in a queueing system

- Objective

$$\text{Min } J(\alpha) = C(\alpha) + cE[D(\alpha)]$$

- Modelling of the objective function

- Case 1: Cost function is known analytically

If $J(\alpha)$ known analytically, α^* is found where $dJ(\alpha)/d\alpha = 0$, i.e.

$$dC(\alpha)/d\alpha + c \, dE[D(\alpha)]/d\alpha = 0$$

This formula is solved if there is a formula for $dE[D(\alpha)]/d\alpha$, e.g. $E[D] = \lambda/(\mu(\alpha)(\mu(\alpha) - \lambda))$ (M/M/1 queue)

- Case 2: Cost function should be estimated

In case of a more complicated model for arrivals and services, we require a Monte Carlo estimate for $dE[D(\alpha)]/d\alpha$

- Solution technique

- Analytical solution via setting $dJ(\alpha)/d\alpha = 0$
- Stochastic approximation $\alpha_{j+1} = \alpha_j - k_j [dC(\alpha)/d\alpha + c \, dE[D(\alpha)]/d\alpha]$

Optimal design of the system

- Example 1: Determine the optimal service speed in a queueing system

➤ Objective:

$$\text{Min } J(\alpha) = C(\alpha) + cE[D(\alpha)]$$

$$\text{Min } J(\mu) = 400\mu + 15 \frac{0.9}{\mu(\mu-0.9)}$$

- Assumption 1: $\alpha = \mu$ (service speed is the only characteristic considered in the service design)
- Assumption 2: $C(\alpha) = 400\mu$ (the cost function is a linear function of the service speed)
- Assumption 3: $E[D(\alpha)] = \frac{\lambda}{\mu(\mu-\lambda)}$ (the waiting time for M/M/1 queue with $c = 15$ (the waiting time cost))
- Assumption 4: $\lambda = 0.9$ (the mean interarrival rate)

➤ Cost function is known analytically

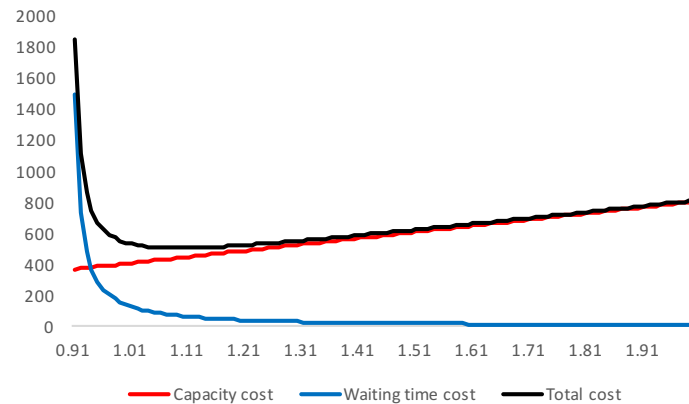
- α^* is found where $\frac{dJ(\alpha)}{d\alpha} = 0$, i.e. $\frac{dJ(\mu)}{d\mu} = 400 - 15 \frac{0.9(2\mu-0.9)}{\mu^2(\mu-0.9)^2} = 0$
- $J(\mu)$ is convex since $\frac{d^2J(\mu)}{d\mu^2} = 15 \times 2 \frac{0.9(3\mu^2 - 3 \times 0.9\mu + 0.9^2)}{\mu^3(\mu-0.9)^3} \geq 0$ as long as $\mu > \lambda$ (necessary condition to reach steady-state for M/M/1)

Optimal design of the system

- Example 1: Determine the optimal service speed in a queueing system

➤ Objective: $\text{Min } J(\mu) = 400\mu + 15 \frac{0.9}{\mu(\mu-0.9)}$

- Graph



- Optimal solution via Stochastic approximation (Newton's method – error margin = 0.001)

Iteration	x_i	$f(x_i)$	$df(x_i)/dx$	$d^2f(x_i)/dx^2$	x_{i+1}	$ x_i - x_{i+1} $
1	1	535	-1085	29970.0000	1.0362	0.0362
2	1.0362	510.1351	-394.6019	11846.0909	1.0695	0.0333
3	1.0695	502.2688	-108.9005	6134.4411	1.0873	0.0178
4	1.0873	501.2103	-15.0452	4544.8588	1.0906	0.0033
5	1.0906	501.1850	-0.3914	4311.1274	1.0907	0.0001

<0.001

Optimal design of the system

- Example 2: Chemical engineering I
 - Consider an investigation into the effect of the concentration of the reactant and the feed rate on the viscosity of product from a chemical process. Let the reactant concentration be factor A, and let the two levels of interest be 15% and 25%. The feed rate is factor B, with the low level being 20 lb/hr (= 20 pound per hour) and the high level being 30 lb/hr. Both factors are quantitative factors.
 - The experiment is replicated four times and the data are as follows:

	Factors		Simulation runs				Avg
	Factor A	Factor B	Run 1	Run 2	Run 3	Run 4	
Design 1	-	-	151	148	159	150	152.00
Design 2	+	-	160	150	160	155	156.25
Design 3	-	+	135	138	141	139	138.25
Design 4	+	+	150	152	146	149	149.25

Optimal design of the system

- Example 2: Chemical engineering I

- Questions

- Estimate the coefficients of the first order regression model ($J(\alpha) = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_{12}x_1x_2$) by calculating the main effects and the interaction effect.
- Interpret the main effects and the interaction effect. Conduct an analysis of variance to determine which factors are significant.
- Optimise the settings of the input factors to maximise the response given the first order model estimated in (a). Assume that only non-negative values are allowed.

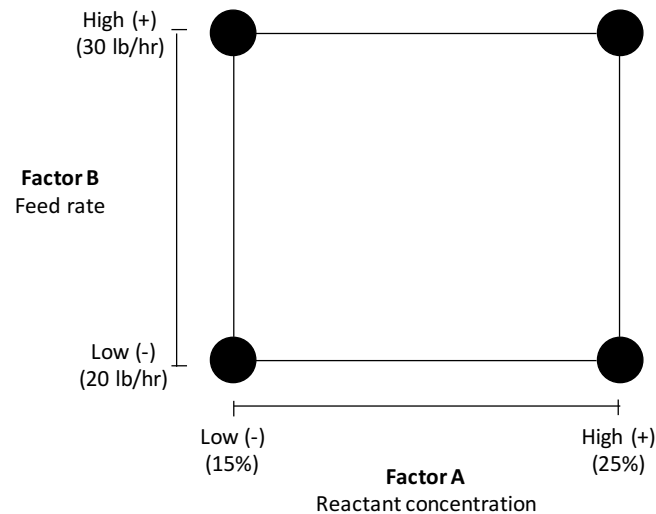
Optimal design of the system

- Example 2: Chemical engineering I

- Preliminaries

- Full factorial design: 2^2 different design points

- Low levels of A and B: -I
- High levels of A and B: +I



Optimal design of the system

- Example 2: Chemical engineering I

➤ Question: Estimate the coefficients of the first order regression model ($J(\alpha) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$) by calculating the main effects and the interaction effect.

➤ Answer:

- The average or main effect of a factor is defined as the change in response produced by a change in the level of that factor averaged over the levels of the other factor.

$$e_A = \frac{(R_2 - R_1) + (R_4 - R_3)}{2} = \frac{-152 + 156.25 - 138.25 + 149.25}{2} = 7.625$$

$$e_B = \frac{(R_3 - R_1) + (R_4 - R_2)}{2} = \frac{-152 - 156.25 + 138.25 + 149.25}{2} = -10.375$$

- Interaction effects

$$e_{AB} = \frac{1}{2} [(R_4 - R_3) - (R_2 - R_1)] = \frac{152 - 156.25 - 138.25 + 149.25}{2} = 3.375$$

Optimal design of the system

- Example 2: Chemical engineering I

➤ Question: Interpret the main effects and the interaction effect. Conduct an analysis of variance to determine which factors are significant.

➤ Answer: Determine variance $S_j^2(n) = \frac{\sum_{i=1}^n (e_j^i - \bar{e}_j(n))^2}{n-1}$ or the standard deviation

$$S_A(n) = 3.683$$

$$S_B(n) = 5.25$$

$$S_{AB}(n) = 1.796$$

	Results per run		
	A	B	AB
Run 1	12	-13	3
Run 2	8	-4	6
Run 3	3	-16	2
Run 4	7.5	-8.5	2.5

Table. Main and interaction effect per run (e_j^i)

- Determine the confidence interval (individual confidence level: 0.95)

Factor A: $\mu_A \in [1.76; 13.48]$ => **Factor A is significant: Increasing A, will increase the response**

Factor B: $\mu_B \in [-18.72; -2.02]$ => **Factor B is significant: Increasing A, will decrease the response**

Factor AB: $\mu_{AB} \in [0.51; 6.23]$ => **Interaction effect AB is significant: Different response if A or B is increased, depending upon the level of the other factor**

Optimal design of the system

- Example 2: Chemical engineering I

- Question: Optimise the settings of the input factors to maximise the response given the first order model estimated in (a). Assume that only non-negative values are allowed.
- Answer: Convert the effect estimates of the 2^2 factorial design into a regression model to predict the response at any point in the space.

Model: $J(\alpha) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$

Factor A (reactant concentration): Coded variable x_1

$$x_1 = \frac{Conc - (Conc_{low} + Conc_{high})/2}{(Conc_{high} - Conc_{low})/2} = \frac{Conc - (15 + 25)/2}{(25 - 15)/2} = \frac{Conc - 20}{5}$$

if $Conc = 25$ then $x_1 = 1$; if $Conc = 15$ then $x_1 = -1$

Factor B (feed rate): Coded variable x_2

$$x_2 = \frac{Feed - (Feed_{low} + Feed_{high})/2}{(Feed_{high} - Feed_{low})/2} = \frac{Feed - (20 + 30)/2}{(30 - 20)/2} = \frac{Feed - 25}{5}$$

if $Feed = 30$ then $x_2 = 1$; if $Feed = 20$ then $x_2 = -1$

Optimal design of the system

- Example 2: Chemical engineering I

- Question: Optimise the settings of the input factors to maximise the response given the first order model estimated in (a). Assume that only non-negative values are allowed.

- Answer: The fitted regression model

$$J(\alpha) = 148.9 + \frac{7.625}{2}x_1 + \frac{-10.375}{2}x_2 + \frac{3.375}{2}x_1x_2$$

$$J(\alpha) = 148.9 + 3.8125x_1 - 5.1875x_2 + 1.6875x_1x_2$$

with $\beta_0 = \bar{Y} = \frac{\sum_{ijk} y_{ijk}}{n \times 2^2}$ (the grand average of all 16 observations)

$\beta_1, \beta_2, \beta_{12} =$ one-half of the factor estimates

(note that the effect estimate is based on a two unit change from -1 to +1)

Optimal design of the system

- Example 2: Chemical engineering I

- Question: Optimise the settings of the input factors to maximise the response given the first order model estimated in (a). Assume that only non-negative values are allowed.

- Answer: The fitted regression model

$$J(\alpha) = 148.9 + \frac{7.625}{2}x_1 + \frac{-10.375}{2}x_2 + \frac{3.375}{2}x_1x_2$$

$$J(\alpha) = 148.9 + 3.8125x_1 - 5.1875x_2 + 1.6875x_1x_2$$

Explanatory power of model: $R^2 = \frac{SS_{model}}{SS_T} = \frac{SS_A + SS_B + SS_{AB}}{SS_T} = \frac{14.5 + 26.9 + 2.8}{59} = 0.751$

with $SS_A = \frac{((R_2 - R_1) + (R_4 - R_3))^2}{n \times 2^2}$ $SS_B = \frac{((R_3 - R_1) + (R_4 - R_2))^2}{n \times 2^2}$ $SS_{AB} = \frac{((R_4 - R_3) - (R_2 - R_1))^2}{n \times 2^2}$

(sum of squares)

$$SS_T = \frac{\sum_{ijk} (y_{ijk} - \bar{Y})^2}{n \times 2^2 - 1} \quad (\text{overall variance in the sample})$$

Optimal design of the system

- Example 2: Chemical engineering I

➤ Question: Optimise the settings of the input factors to maximise the response given the first order model estimated in (a). Assume that only non-negative values are allowed.

➤ Answer: Optimise the fitted regression model

$$J(\alpha) = 148.9 + 3.8125x_1 - 5.1875x_2 + 1.6875x_1x_2$$

First derivative (x_1): $J'(\alpha, x_1) = \frac{dJ(\alpha)}{dx_1} = 3.8125 + 1.6875x_2$ $J'(\alpha, x_1) = 0$ if $x_2 = -\frac{3.8125}{1.6875} = -2.259$

or if you convert this coded variable to the real independent variable: $x_2 = \frac{Feed-25}{5}$

=> Feed rate = 13.703 lb/hr

First derivative (x_2): $J'(\alpha, x_2) = \frac{dJ(\alpha)}{dx_2} = -5.1875 + 1.6875x_1$ $J'(\alpha, x_2) = 0$ if $x_1 = \frac{5.1875}{1.6875} = 3.074$

or if you convert this coded variable to the real independent variable: $x_1 = \frac{Conc-20}{5}$

=> Concentration = 35.370%

Objective function value: $J(\alpha) = 148.9 + 3.8125 \times (3.074) - 5.1875 \times (-2.259) + 1.6875 \times (3.074) \times (-2.259)$

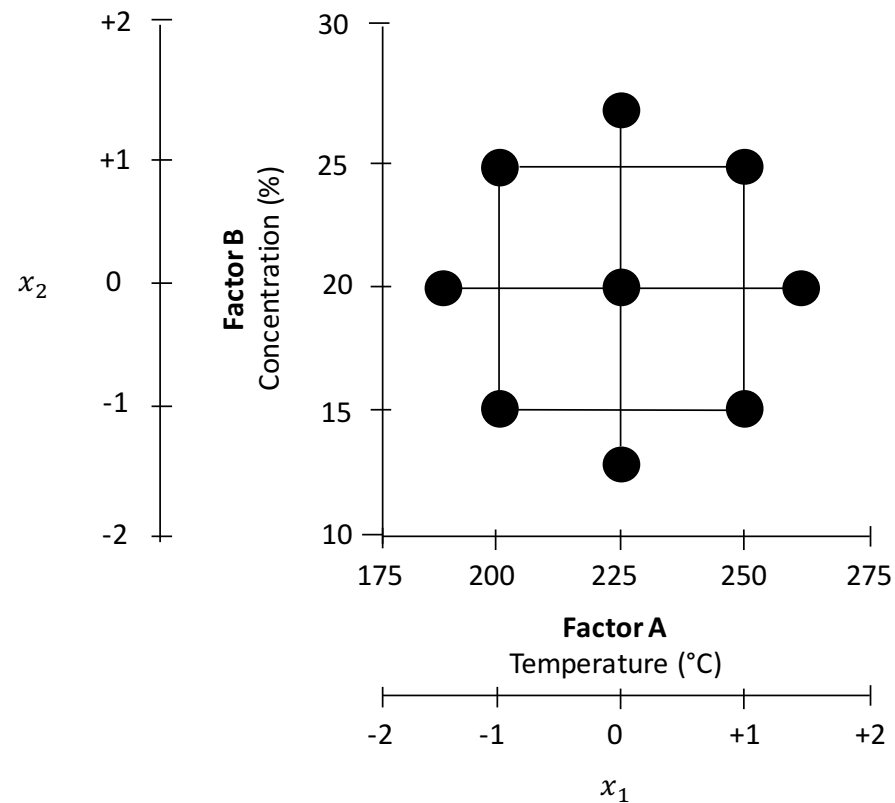
$$J(\alpha) = 160.657$$

Optimal design of the system

- Example 3: Chemical engineering II
 - Consider an investigation into the effect of the temperature (ξ_1) and the concentration of the reactant (ξ_2) on the percentage conversion of a chemical process (y). Both factors are quantitative factors. The chemical engineer has used an approach to improve this process based on designed experiments. The first experiment was a screening experiment involving several factors that isolated the temperature and the concentration as the two most important variables. Because the experimenters thought that the process was operating in the vicinity of the optimum they elected to fit a quadratic model relating to yield to temperature and concentration.
 - Question: Optimise the settings of the input factors to maximise the response using a second order model.

Optimal design of the system

- Example 3: Chemical engineering II
 - The design of experiments is a central composite design



Optimal design of the system

- Example 3: Chemical engineering II
 - Central composite design: Coded variables and real variables

Factor A (Temperature): Coded variable x_1

$$x_1 = \frac{Temp - (Temp_{low} + Temp_{high})/2}{(Temp_{high} - Temp_{low})/2} = \frac{Temp - (200 + 250)/2}{(250 - 200)/2} = \frac{Temp - 225}{25}$$

if $Temp = 250$ then $x_1 = 1$; if $Temp = 200$ then $x_1 = -1$; if $x_1 = -1.414$ then $Temp = 189.65$; ...

Factor B (Concentration): Coded variable x_2

$$x_2 = \frac{Conc - (Conc_{low} + Conc_{high})/2}{(Conc_{high} - Conc_{low})/2} = \frac{Conc - (15 + 25)/2}{(25 - 15)/2} = \frac{Conc - 20}{5}$$

if $Conc = 25$ then $x_2 = 1$; if $Conc = 15$ then $x_2 = -1$; if $x_2 = -1.414$ then $Conc = 12.93$; ...

Optimal design of the system

- Example 3: Chemical engineering II
 - Central composite design: Simulation results

Observation	x_1	x_2	Temperature (°C)	Conc. (%)	y
1	-1	-1	200	15	43
2	1	-1	250	15	78
3	-1	1	200	25	69
4	1	1	250	25	73
5	-1.414	0	189.65	20	48
6	1.414	0	260.35	20	78
7	0	-1.414	225	12.93	65
8	0	1.414	225	27.07	74
9	0	0	225	20	76
10	0	0	225	20	79
11	0	0	225	20	83
12	0	0	225	20	81

Optimal design of the system

- Example 3: Chemical engineering II

- Convert the effect estimates of the composite design into a regression model to predict the response at any point in the space.

Model: $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2$

Model: $\hat{y} = 79.75 + 10.18x_1 + 4.22x_2 - 8.50x_1^2 - 5.25x_2^2 - 7.75x_1x_2$ (coded variables)

Model: $\hat{y} = -1080.22 + 7.767\xi_1 + 23.1932\xi_2 - 0.0136\xi_1^2 - 0.21\xi_2^2 - 0.062\xi_1\xi_2$ (real variables)

=> According to the analysis of variance are all coefficients significant (linear/interaction/quadratic effects)

Optimal design of the system

- Example 3: Chemical engineering II

- Optimise the fitted regression model

Model: $\hat{y} = 79.75 + 10.18x_1 + 4.22x_2 - 8.50x_1^2 - 5.25x_2^2 - 7.75x_1x_2$ (coded variables)

$$\text{First derivative } (x_1): \hat{y}'(\alpha, x_1) = \frac{d\hat{y}}{dx_1} = 10.18 - 17x_1 - 7.75x_2$$

$$\text{First derivative } (x_2): \hat{y}'(\alpha, x_2) = \frac{d\hat{y}}{dx_2} = 4.22 - 10.5x_2 - 7.75x_1$$

Conditions for optimality:

$$\hat{y}'(\alpha, x_1) = 0$$

$$\hat{y}'(\alpha, x_2) = 0 \quad \Rightarrow 2 \text{ equations, 2 unknown variables}$$

$$\Rightarrow x_1 = 0.6263 \quad \text{or Temperature} = 240.65^\circ\text{C}$$

$$\Rightarrow x_2 = -0.0604 \quad \text{or Concentration} = 19.69\%$$

Optimal design of the system

- Alternative Heuristic optimisation: Simulated Annealing

- What if Stochastic approximation is not appropriate?

- $J(\alpha)$ is not convex and there are many local minima

- => Stochastic approximation will only find the local minimum depending on α_0 , not the global minimum

- When the error in the regression model is too large and no adequate fitting is possible

- => The curve is not known, estimate the first derivative in order to go towards the optimum

- $\alpha \in \mathcal{A}$ is discrete

- => The derivatives of $J(\alpha)$ have no meaning

- Use Heuristic optimisation: E.g. Simulated annealing

- What? A *heuristic* is a procedure that provides good solutions within reasonable times, but for which optimality cannot be guaranteed.

- When? When optimal procedures are not useful.

- How? Developing a good heuristic requires *problem-specific* insight, creativity and experience.

Optimal design of the system

- Alternative Heuristic optimisation: Simulated Annealing

- Basic Concepts of Heuristic optimisation

- Representation: The encoding of candidate solutions implies the search space and its size.
- Evaluation function
 - Returns a specific value that indicates the quality of any particular solution.
 - A mapping from the space of possible candidate solutions to a set of numeric values that indicate the solution quality.
- Search procedure
 - Search strategy to go from solution to solution is highly dependent on the philosophy of the (meta-)heuristic itself (e.g. Simulated annealing)
 - Conceptual frameworks that guide and modify the operations of subordinate heuristics and combine different learning strategies and structured information in order to explore the solution space intelligently and produce high-quality solutions efficiently
 - Dynamic balance between local improvement procedures (*intensification*) and higher level search strategies (*diversification*):
 - It is important to quickly identify regions in the search space with high quality solutions.
 - The search mechanism should avoid to waste too much time in regions of the search space which are either already explored or which do not provide high quality solutions.

Optimal design of the system

- Alternative Heuristic optimisation: Simulated Annealing

- Basic Concepts of Simulated Annealing

- Inspired by Local-search based Improvement heuristics (search learning strategy)

- What? Local search methods comprehend an intensive exploration of the solution space that is performed by moving at each step from the current (complete) solution to another promising solution in its neighborhood.

- How to make a move in the solution space? Neighbourhood concept

- Neighbourhood of a schedule = The set of solutions that can be reached after a simple move operation

- Small vs large neighbourhoods

- Pseudocode

- do

- local = 0

- Evaluate the fitness of the solution element $v_c (= f(v_c))$

- Select all new points in the neighbourhood of v_c

- Select the point v_n from the set of new points with the best fitness value $f(v_n)$

- if ($f(v_n) < f(v_c)$)

- Update solution element $v_n \rightarrow v_c$

- local = 1

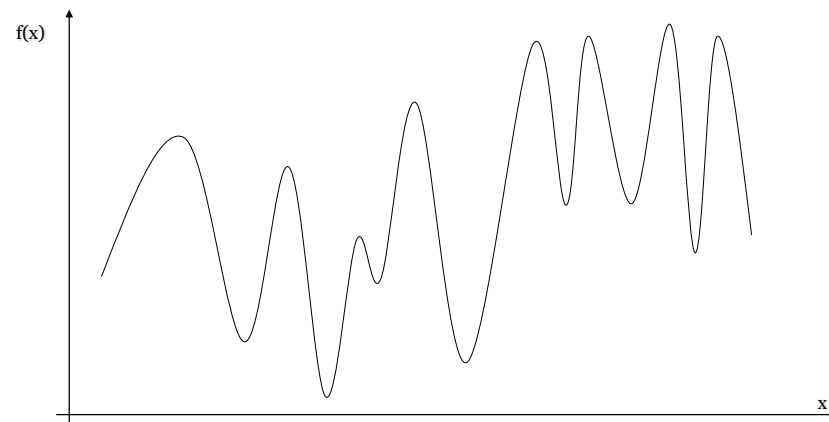
- while (local = 1)

Optimal design of the system

- Alternative Heuristic optimisation: Simulated Annealing

- Basic Concepts of Simulated Annealing

- Inspired by Local-search based Improvement heuristics (search learning strategy)
 - Objective function $f(x)$ (minimization problem): Solution space: 2-dimensional example

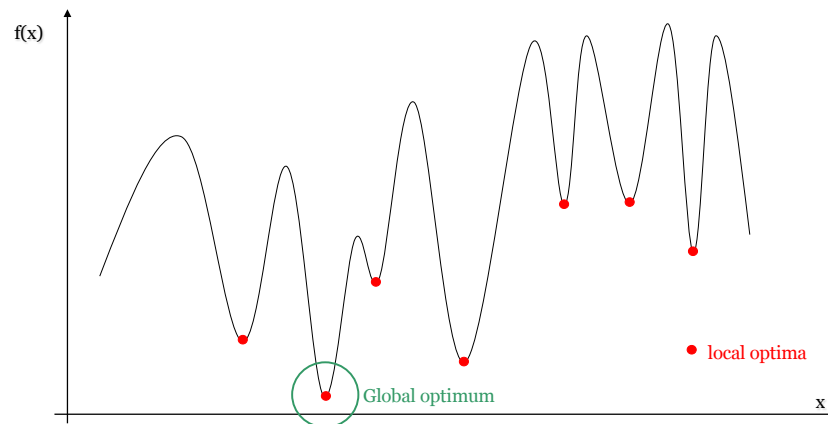
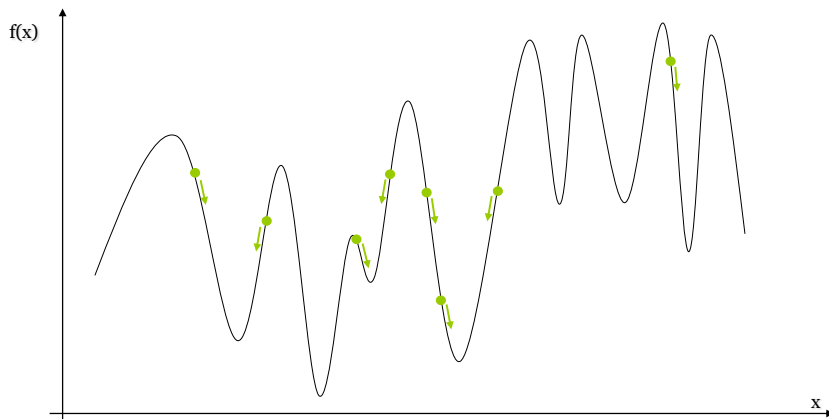


Optimal design of the system

- Alternative Heuristic optimisation: Simulated Annealing

- Basic Concepts of Simulated Annealing

- Inspired by Local-search based Improvement heuristics (search learning strategy)
 - Move towards best solution in the neighbourhood
 - Results in many local optima and 1 global optimum



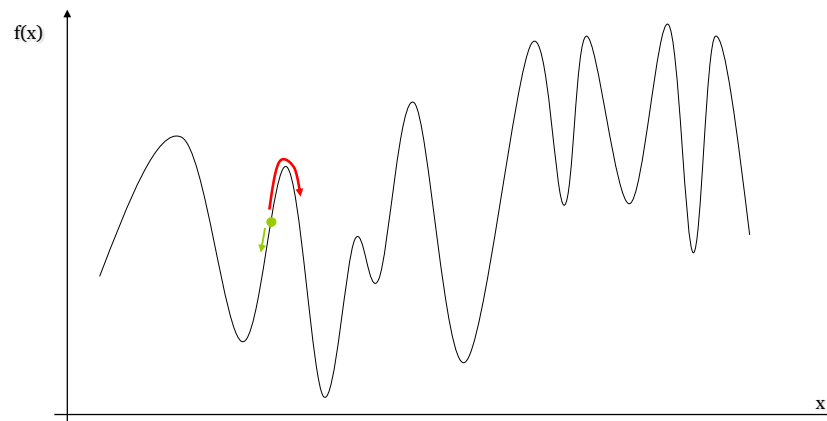
Optimal design of the system

- Alternative Heuristic optimisation: Simulated Annealing

- Basic Concepts of Simulated Annealing

- Escape local optima

- After applying an improvement heuristic, a local optimum in the solution space is reached. Various techniques are available for escaping local optima, after which the improvement heuristic will lead to other local optima. Simulated annealing is an iterative search method that will sometimes climb in a hill in order to reach other valleys
- Diversification in search: Worsening moves are accepted with a probability that decreases with the degree solution quality deteriorates and with increasing running time.



Optimal design of the system

- Alternative Heuristic optimisation: Simulated Annealing

- Basic Concepts of Simulated Annealing

- Iterative search: $\alpha_0 \rightarrow \alpha_1 \rightarrow \alpha_2 \rightarrow \dots \rightarrow \alpha^*$
 - Sometimes in good direction: $J(\alpha_{i+1}) < J(\alpha_i)$, i.e. downhill towards local minimum
 - Sometimes in wrong direction: $J(\alpha_{i+1}) > J(\alpha_i)$, i.e. uphill towards other valley
- Temperature
 - Define a temperature T
 - The higher the temperature, the more likely a step in the wrong direction will be taken
- Neighbourhood
 - For every $\alpha \in \mathcal{A}$, we can define a set of neighbouring values $\mathcal{N}(\alpha)$
 - SA will only take steps towards neighbouring values

Optimal design of the system

- Alternative Heuristic optimisation: Simulated Annealing

- Basic Concepts of Simulated Annealing

- SA Solution update $\alpha_i \rightarrow \alpha_{i+1}$
 - Propose a candidate neighbour $\tilde{\alpha}_{i+1} \in \mathcal{N}(\alpha_i)$
 - Estimate cost difference: $\bar{\tau}_i = \bar{J}(\tilde{\alpha}_{i+1}) - \bar{J}(\alpha_i)$
 - If $\bar{\tau}_i < 0$ then candidate goes into good direction
=> always accept it, i.e. with probability $p_i^{acc} = 1$
 - If $\bar{\tau}_i > 0$ then candidate goes into wrong direction
=> accept it with probability $p_i^{acc} = e^{-\tau_i/T}$

So accept proposed candidate with probability $p_i^{acc} = \min(1, e^{-\tau_i/T})$: $\alpha_{i+1} = \tilde{\alpha}_{i+1}$

- Cooling scheme
 - SA with fixed temperature T results in a random walk through $\mathcal{A}$ as it will never converge
 - Temperature should be lowered in each step: $T_i \rightarrow 0$

Optimal design of the system

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- Pseudocode

```
Begin
  Generate the initial state q
   $T = T_{\max}$ 
  Let  $E(q, T)$  be the associated energy
  while  $(T \geq T_{\min})$ 
    for  $i = 1$  to  $k$ 
      Perturb  $q$  to yield  $s$ 
      Let  $E(s, T)$  be the associated energy
      Set  $q \leftarrow s$  with probability  $1/(1 + e^{-(E(q, T) - E(s, T))/T})$ 
    end for
     $T = rT$ 
  end while
  Decode  $q$  to provide the solution of the problem.
End
```

Reference

S. Bandyopadhyay and S. Saha, 2013, Unsupervised Classification, Springer-Verlag Berlin Heidelberg